The Admissibility-Capacity Ledger

Cosmological Absolute Predictions from Finite Information Capacity

Paper 8 of the Admissibility Physics Framework

E.S. Brooke*

April 2026 | Version 2.5 (Phase 14f $\mathrm{I_k}$-stratification: two bridges + two composed self-identifications)

Abstract

The Admissibility Physics Framework (APF) derives the structural content of the Standard Model — gauge group $SU(3) \times SU(2) \times U(1)$, fermion generations $N_{\mathrm{gen}} = 3$, the weak mixing angle $\sin^2\theta_W = 3/13$, and other dimensionless observables — from a single axiom, **A1** (finite information capacity), through the variational *Principle of Least Enforcement Cost* (**PLEC**). APF predicts dimensionless structural ratios in Planck units; comparison with laboratory measurements in SI units requires only the standard conversion from Planck to SI, exactly as quantum field theory and lattice QCD require when comparing their own natural-units predictions to experiment. This paper applies the ledger machinery built in *Paper 1* (*The Enforceability of Distinction*) through *Paper 7* (*A Minimal Quantum of Action*) to the cosmological sector. The result is a compact, zero-free-parameter prediction stack: the three observed cosmological residual fractions are proven to be information-entropy allocations on substructures of the Standard-Model interface ($\Omega_b = 3/61 = S(V_b)/S_{\mathrm{SM}}$, $\Omega_c = 16/61 = S(V_c)/S_{\mathrm{SM}}$, $\Omega_\Lambda = 42/61 = S(V_\Lambda)/S_{\mathrm{SM}}$, with V_b, V_c, V_Λ structurally defined subspaces of the SM slot space V_{61}); the dark-energy density predicts $\rho_\Lambda/M_{\mathrm{Pl}}^4 = 42/102^{62}$ in Planck units; the algebraically-linked Hubble constant predicts $H_0/M_{\mathrm{Pl}} = \sqrt{8\pi K_{\mathrm{SM}}/3}/102^{31}$, corresponding to $H_0 \approx 70.03\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ near the present Hubble-tension midpoint between Planck 2018 [1] and SH0ES 2022 [2]; and the CMB temperature satisfies an independent formula $(T_{\mathrm{CMB}}/M_{\mathrm{Pl}})^4 = 48/102^{64}$ at 0.33% of the FIRAS measurement [3], with the species-dependent exponent difference $\Delta k = 2$ matching the photon polarization count. The paper's new structural content is the *three-level identity stack* at each T_ACC consistency identity (integer, scalar, and subspace witnesses, all proven [P] by the constructions of §5) together with an *operator-level closure* constructing the thermal density matrix on an explicit finite-dimensional microstate Hilbert space and recovering the cosmological formulas as $\beta \to 0$ limits. The four identities close by two distinct theorem species: I_2 (always) and I_1 (at the canonical SM-dS joint point $K = 42$) carry *bridge theorems* with three independent constructions each converging on a common 42-dimensional subspace; I_3 and I_4 carry *operator-level composed self-identifications* with three and four constructions respectively (the I_3/I_4 closure is single-interface rather than cross-interface, because the quantum-regime slot scale $K = 1$ collapses the bridge route by construction). A dual-convention formulation of the horizon interface ($d_{\mathrm{eff}} = e$ for ACC bookkeeping, $d_{\mathrm{eff}} = 2$ for literal qubit-tensor-product microstate realisation) formalises the I_1 joint-point bridge's Convention B construction. The cosmological predictions follow from this structural layer as direct applications of the ledger machinery.

*brooke.ethan@gmail.com; ORCID 0009-0001-2261-4682.

Scope note (Lane A). Paper 8 is a *formal framework paper*. Its principal content is the ledger architecture (the ACC record, the six regime projections, the four consistency identities) and the bridge theorem at I_2; the cosmological-sector formulas are presented as structural consequences of that architecture, not as a cosmological-inference fit. The horizon identity I_1 uses the Bekenstein cell-count convention $K_{\text{horizon}} = A/(4\ell_P^2)$, which reproduces the standard Bekenstein–Hawking entropy $S_{\text{BH}} = A/(4\ell_P^2)$ exactly; no prefactor mutation is proposed.

Integrated non-trivial content (Supplement §O). The paper's non-trivial structural mechanism — the I_2 bridge, its gauge-uniqueness strengthening, its framework-internal forcing chain yielding $(K_{\text{SM}}, d_{\text{eff}}^{\text{SM}}) = (61, 102)$, its structurally-mandatory two-tier architecture, and its brittle cosmological consequences — is collected in the centerpiece Supplement §O. Four strengthened-in-v2.3 mathematical results appear there in compressed form: Theorem 7.14′ (I_2 algebraic closure: Type III_1 factor at infinite volume via Araki–Connes classification); Theorem C.5 (single-scale impossibility: the two-tier architecture is structurally mandatory); Theorem 5.7 (V_Λ gauge-uniqueness: the 42-dim bridge subspace is the unique G_{SM}-invariant complement of V_{local}); and a full-covariance Planck 2018 multivariate Bayes-factor analysis over $(\Omega_b, \Omega_c, \Omega_\Lambda)$ giving $\chi^2 = 1.18$ ($p \approx 0.76$, Mahalanobis distance 1.09σ). A competitive statistical comparison to ΛCDM posteriors under full Planck + BAO + SN likelihoods with nuisance parameters is explicitly deferred to a companion phenomenology paper. Section 9's operator realisation is finite-dimensional, with an infinite-volume extension available via the Bratteli–Robinson inductive limit (Supplement §H.5.2, Theorem 7.14′); no algebraic-QSM content beyond standard machinery is claimed.

1. Introduction

The Admissibility Physics Framework (APF) is built on one assumption: *the universe cannot maintain infinitely many enforceable distinctions with finite resource.* Rigorously stated as Axiom **A1** (Finite Enforcement Capacity) and combined with the variational *Principle of Least Enforcement Cost* (**PLEC**), this finiteness assumption is sufficient to derive the structural content of the Standard Model — gauge group $SU(3) \times SU(2) \times U(1)$, fermion generation count $N_{\text{gen}} = 3$, weak mixing angle $\sin^2\theta_W = 3/13$, and the cosmological density partition $(\Omega_b, \Omega_c, \Omega_\Lambda) = (3/61,\ 16/61,\ 42/61)$ — as accounting theorems on a finite budget. Every prediction is a closed-form rational number or a closed-form expression in rational numbers and the Planck mass; no parameter is tuned. This paper applies the machinery to the cosmological sector specifically and derives the absolute values of $\rho_\Lambda, H_0, T_{\text{CMB}}$, together with the full Standard-Model entropy dictionary.

Hierarchy of claims (read this before anything else). Paper 8 contains four regime-consistency identities I_1, I_2, I_3, I_4. *Three of them are intentionally modest*: I_1 is a horizon-convention identity under the Bekenstein cell-count convention, reproducing the standard $S_{\text{BH}} = A/(4\ell_P^2)$ by construction; I_3 is a standard finite-dimensional entropy closure $S_{\text{vN}}(\rho_{\max}) = \ln N$; I_4 is a standard high-temperature partition-function limit $\lim_{\beta\to 0} \ln Z = \ln N$ on a finite admissible set. The paper's only non-trivial structural claim is the *gauge-cosmological bridge* I_2: a unique 42-dimensional G_{SM}-invariant subspace $V_\Lambda \subset V_{61}$ exists under the stated G_{SM}-action and T12 partition constraint, inducing the residual partition $3 + 16 + 42 = 61$ and the cosmological fraction $\Omega_\Lambda = 42/61$. **This theorem is stated in fully self-contained form as Theorem 1.1 of the Technical Supplement §1 ("Formal Kernel of the Paper"), placed immediately after the supplement's**

reader map for exactly this reason. A reader wanting the structural headline of Paper 8 should read Supplement §1 first; the integrated centerpiece reading is Supplement §O, and a fully explicit minimal working example at toy-scale ($K = 3$, $d_{\text{eff}} = 4$) is Supplement §R.

1.1 Three primitives — distinction, enforcement, capacity

APF's atomic vocabulary is three terms, each with a specific technical meaning. Everything the framework derives unpacks from these three.

A distinction d is a maintained differentiation: the assertion, backed by a physical mechanism, that "this is not that." An electron is distinguished from a muon by a mass hierarchy enforced through Yukawa couplings. The spin-up outcome of a measurement is distinguished from spin-down by the orthogonality of the outcome projectors. The interior of a de Sitter horizon is distinguished from its exterior by the finite area that bounds them. Without some physical mechanism actively separating the two alternatives, the distinction does not exist.

Enforcement cost $\varepsilon(d)$ is the structural resource required to keep distinction d intact against alternatives. Exact symmetries — the $SU(3)$ colour symmetry that routes the strong force, the $U(1)_{\text{em}}$ symmetry that conserves electric charge — are *free* ($\varepsilon = 0$): they are maintained by algebraic identity and require no ongoing expenditure. Broken symmetries are not free: the weak force, post-electroweak breaking, carries a cost $\varepsilon \propto v = 246$ GeV. Spectral-gap distinctions (the electron mass, the neutrino mass, the muon-electron split) cost proportionally to the gap. Topological distinctions (colour confinement) cost through the string tension $\kappa \approx 1$ GeV/fm. APF does not posit a specific functional form for $\varepsilon(d)$; it posits only that the cost is positive on non-trivial distinctions and additive across independent ones.

Capacity $C(\Gamma)$ is the total enforcement budget at a specified physical interface $\Gamma = (\rho, \Gamma)$ — the pair of a state and the resource structure on which physics is described. The central claim of APF is that this budget is finite: $C(\Gamma) < \infty$ at every interface. Nothing in the framework posits a specific numerical value for C; only that the supremum over enforceable distinctions is finite.

1.2 Axiom A1 and the Principle of Least Enforcement Cost

From these three primitives, APF posits a single axiom and a single variational principle.

Axiom A1 — Finite Enforcement Capacity

At every physical interface $\Gamma = (\rho, \Gamma)$, the total enforcement cost across maintained distinctions is bounded by a finite capacity:

$$\sum_{d \in \mathcal{D}} \varepsilon(d) \;\leq\; C(\Gamma) \;<\; \infty,$$

where $\mathcal{D}$ is the set of maintained distinctions, $\varepsilon(d) > 0$ is the enforcement cost per distinction, and $C(\Gamma) > 0$ is the interface's capacity.

A1 does two things at once: it carves out the set of *admissible* configurations — those for which $\sum_d \varepsilon(d) \leq C$ — and it declares that set bounded. Call it the *budget window*, $\text{BW}(\Gamma)$. Every

maintainable physical configuration lies inside BW(Γ). Which configuration is actually *realised*? That is the second half of the framework:

> **Principle of Least Enforcement Cost (PLEC)**
> The realised physical configuration at interface Γ is the minimiser of $\sum_d \varepsilon(d)$ over the budget window BW(Γ), subject to a minimum-distinction floor $\varepsilon^* > 0$ (the least enforceable distinction the interface supports).

PLEC has the same structural form as other variational principles in physics. Lagrangian mechanics selects trajectories that extremise $\int L\,dt$; Fermat's principle selects light paths that extremise optical length; the Gibbs principle selects equilibrium configurations that extremise free energy. PLEC selects physical configurations that extremise enforcement cost. What distinguishes PLEC from these classical variational principles is the *quantity being extremised*: it is information-theoretic weight on the count of enforceable distinctions, not spacetime action or free energy. Every prediction in this paper, and every prediction in Papers 1 through 7, follows from **A1** and PLEC.

1.3 What the framework derives — a taster

To give the reader immediate grip on what "derives" means in APF, Table 1 compiles selected outputs across the paper series. Every APF entry is a closed-form rational number or a closed-form expression in rational numbers and the Planck mass $M_{\rm Pl} \equiv \sqrt{\hbar c/G}$. No entry is tuned; every entry is a theorem.

Quantity	**APF form**	**APF value**	**Observed (source)**	**Match**	**Paper**
Gauge group	$SU(3)\times SU(2)\times U(1)$	(structural)	SM (experiment)	exact	4
Fermion generations	$N_{\rm gen} = 3$	3	3 (PDG [4])	exact	2
SM capacity	$K_{\rm SM} = 61$	61	$12 + 45 + 4 = 61$ (SM d.o.f.)	exact	2
Weak mixing angle	$\sin^2\theta_W = 3/13$	0.2308	0.23121 $\pm$ 0.00004 [4]	0.18%	4 + [5]
Baryon density	$\Omega_b = 3/61$	0.04918	0.0493 ± 0.0006 [1]	0.4%	8
Dark-matter density	$\Omega_c = 16/61$	0.2623	0.261 ± 0.002 [1]	0.6%	8
Dark-energy density	$\Omega_\Lambda = 42/61$	0.6885	0.6889 ± 0.0056 [1]	0.06%	8
CMB temperature	$(T_{\rm CMB}/M_{\rm Pl})^4 = 48/102^{64}$	2.717 K	2.7255 $\pm$ 0.0006 K [3]	0.33%	8
Dark-energy absolute	$\rho_\Lambda/M_{\rm Pl}^4 = 42/102^{62}$	$10^{-122.91}$	$10^{-122.94\pm0.02}$ [1, 2]	~8%	8
Hubble constant	$H_0/M_{\rm Pl} = \sqrt{8\pi K/3}/102^{31}$	70.03 km/s/Mpc	67.4/73.0 [1, 2]	tension midpoint	8

Table 1: Selected derived structural outputs of APF. The *APF form* column gives the closed-form structural expression; the *APF value* column gives the dimensionless or Planck-unit-converted numerical value; the *observed* column gives the present best measurement; the *match* column gives the fractional agreement. Every entry is derived from Axiom **A1** and PLEC through the constructions of Papers 1 through 8; no entry is a fit. The Hubble-tension row marks the known observational controversy between Planck 2018 and SH0ES 2022 whose size coincides with the APF prediction's apparent deviation on $\rho_\Lambda/M_{\rm Pl}^4$; the algebraically-linked H_0 row shows APF landing at the midpoint of that tension. Paper-by-paper delivery of each row is summarised in Table 2.

The table's claim is bold: all ten of these dimensionless structural quantities are derived from a single axiom and a single variational principle. In every case where an observational comparison is available, the APF prediction agrees with measurement to sub-percent precision, or lands at the precise midpoint of a documented observational tension. This paper develops the structural theorems that force the cosmological-sector block (rows 5–10) and brings the full conversion machinery to the surface.

1.4 The paper series — what Papers 1 through 7 deliver

Before stating Paper 8's contribution, it helps to see where in the series each piece of machinery lives. Readers new to APF can treat Papers 1 through 7 as a linear construction; each builds on the machinery of its predecessors, and Paper 8 assumes their outputs as established theorems. Table 2 is the reader's map.

Paper	Title (abbreviated)	Content delivered
1	Enforceability of Distinction	Axiom **A1**; admissibility structure; quantum skeleton; Born rule via Gleason closure
2	Finite Admissibility	Structural capacity $K_{\mathrm{SM}} = 61$; partition $45 + 4 + 12$; generation count $N_{\mathrm{gen}} = 3$
3	Entropy, Time, Accumulated Cost	ACC record (K, d_{eff}); six regime projections; four consistency identities I_1–I_4
4	Admissibility Constraints	Gauge group $SU(3) \times SU(2) \times U(1)$ (1-of-4680 uniqueness); $\sin^2\theta_W = 3/13$
5	Quantum Structure	Quantum projection π_Q; thermo–quantum consistency identity I_3
6	Dynamics and Geometry	Einstein equations from ledger; $V_\Lambda \subset V_{61}$; interface-sector bridge theorem
7	Minimal Quantum of Action	Action–thermo identity I_4; spectral-action equivalence at finite temperature
13	Minimal Admissibility Core	Master reference; code–paper map; every theorem anchored to a bank-registered check

Table 2: The APF paper series. Papers 1 through 7 deliver the ledger machinery and the sector-specific derived outputs of Table 1; Paper 13 is the master reference that anchors every theorem to a named check in the verification bank. Paper 8 (this paper) applies the machinery to the cosmological sector.

1.5 Natural coordinates — Planck units as APF's native language

Because the primitives of APF are counts of maintained distinctions and information-theoretic weights on those counts, the framework's outputs are intrinsically *dimensionless*: integer capacities ($K_{\mathrm{SM}} = 61$), integer partitions ($45 + 4 + 12$), rational ratios of integers ($\sin^2\theta_W = 3/13$), and dimensionless expressions built from these. Neither $\varepsilon(d)$ nor $C(\Gamma)$ has intrinsic units of length, time, mass, or energy — both are structural weights on a count.

Natural coordinates. Natural coordinates for such a framework set the fundamental constants of gravitation, quantum mechanics, and relativity to unity. The *Planck mass*

$$M_{\mathrm{Pl}} \equiv \sqrt{\frac{\hbar c}{G}} \approx 1.22091 \times 10^{19}\ \mathrm{GeV}$$

is the unique energy scale constructible from $\hbar$, c, and Newton's gravitational constant G. APF works in units where $M_{\rm Pl} = 1$, $\hbar = 1$, $c = 1$ throughout. In this coordinate system, every APF prediction takes one of two forms: a pure dimensionless number ($\sin^2\theta_W = 3/13$, $\Omega_\Lambda = 42/61$), or a dimensionless ratio multiplied by an integer power of $M_{\rm Pl}$ ($\rho_\Lambda/M_{\rm Pl}^4 = 42/102^{62}$ for an energy density, $H_0/M_{\rm Pl} = \sqrt{8\pi K/3}/102^{31}$ for an inverse length). In natural units the $M_{\rm Pl}^n$ factor is unity; it re-enters only on conversion to SI for comparison with experiment.

SI conversion. Laboratory measurements report in SI units. To convert any APF prediction to SI, multiply the APF dimensionless ratio by $(M_{\rm Pl}^{\rm SI})^n$ for the prediction's appropriate n, using the experimentally-measured Newton's G at the comparison stage. This is the standard unit-interface practice used by quantum field theory when converting natural-units amplitudes to laboratory cross-sections, by lattice QCD when converting dimensionless hadronic observables to GeV, and by every other natural-units framework. APF imports no dimensional scale at any derivational step; G enters only at the observational-comparison stage, where it is an external measurement rather than a tunable parameter. The foundational development of this framing appears in *Paper 0* (*What Physics Permits*), §"**A1** as a statement about information, not energy," and *Paper 13* (*The Minimal Admissibility Core*), §"Planck Units as Native Convention."

1.6 What this paper contributes

Paper 8 has three main contributions on top of the Table 1 cosmological-sector predictions.

(i) The three-level identity stack. Paper 3 introduced the four consistency identities I_1–I_4 that relate the six regime projections of the ACC record. This paper proves each identity at *three* distinct levels — integer, scalar, and subspace — and establishes that the three levels commute under explicit inter-level functors. At the Standard-Model interface, the headline identity I_2 (gauge–cosmological) is proven fully [P] at all three levels, with the subspace-level witness being the 42-dimensional global-interface stratum $V_\Lambda = V_{\rm global}$ of Paper 6. The remaining identities I_1, I_3, I_4 are closed [P] at their subspace levels by three new constructions (subspace functors) delivered in §5.

(ii) The sector conversion matrix. The six regime projections are not independent quantities — they are sector-specific readings of a single PLEC-selected point, and the four identities I_1–I_4 (together with their compositions) force the full pairwise conversion table between them. Section 7 assembles this into a 6×6 matrix, lists the 15 pairwise structural conversions, and cross-checks each one against observation at the Standard-Model interface.

(iii) The cosmological absolute predictions. Sections 8 and 9 derive the Standard-Model-interface absolute values of $\rho_\Lambda, \rho_b, \rho_c, H_0$, and $T_{\rm CMB}$ from the ledger machinery, together with a species-dependent exponent structure that distinguishes bosonic from polarised-bosonic sectors. The derivation is constructive on an explicit finite-dimensional microstate Hilbert space $H_{\rm micro} = (\mathbb{C}^{d_{\rm eff}})^{\otimes K}$; the cosmological absolute formulas emerge as high-temperature limits of a thermal density matrix on that space.

Epistemic boundaries. It is worth naming plainly what APF does and does not assume. No physical structure is imported into the derivation: the gauge group is derived (Paper 4), not assumed; the generation count is derived (Paper 2), not assumed; cosmological observables are derived in Planck units from the ledger machinery, not fitted. Mathematical theorems used in closure and representation arguments — Gleason's theorem for Born-rule reconstruction, the Stinespring dilation

for completely-positive maps, the Lieb–Ruskai monotonicity for the entropy identity, and others — are cited at point of use as explicit mathematical imports, not hidden physics assumptions. SI conversion uses the experimentally-measured G at the comparison stage only, not at any derivation step.

The remainder of this paper is organised as follows.

Section	**Content**
§2	ACC record (K, d_{eff}); six regime projections; four consistency identities I_1–I_4; T_ACC unification theorem; relation to PLEC
§3	Two-tier carrier structure: slot space V_{slot} and microstate space H_{micro}
§4	Three-level identity stack: integer / scalar / subspace witnesses for each I_k
§5	Four subspace functors closing the subspace-level witness: F_{horizon}, bridge, F_{quantum}, F_{operator}
§6	Fractional Reading Equivalence; immediate consequence: the SM entropy-allocation dictionary
§7	Sector conversion matrix: the 6×6 inter-projection table with dataset cross-check at the SM interface
§8	Cosmological absolute predictions: $\rho_\Lambda, \rho_{\text{crit}}, \rho_b, \rho_c, T_{\text{CMB}}, H_0$; coefficient structural derivation; Hubble-tension placement
§9	Operator-level realisation of the ledger on an explicit finite-dimensional H_{micro}
§10	Scope of the prediction formula; species-dependent exponent hypothesis; what is covered, what is not
§11	Formal falsifiers
§12	Observer-dependence question of the bridge theorem (single substantive open structural question)
§13	Conclusion

Table 3: Section map for Paper 8. The Technical Supplement contains full proofs of every theorem stated in the main paper.

2. The Admissibility-Capacity Ledger

The machinery this paper applies to the cosmological sector is the Admissibility-Capacity Ledger, developed in *Paper 3* (*Entropy, Time, and Accumulated Cost*) and bank-anchored in *Paper 13* (*The Minimal Admissibility Core*). This section restates the machinery in self-contained form: the record, the projections, and the four consistency identities. Paper-8 arguments in later sections import no claim not established here.

2.1 The ACC record

An interface in APF is specified by a state ρ and a resource structure Γ. The admissibility-capacity of the interface (ρ, Γ) is summarized by a single structural record:

Definition — Admissibility-Capacity Record $\mathrm{ACC}(\rho, \Gamma)$

For an interface (ρ, Γ), the admissibility-capacity record is the pair

$$\mathrm{ACC}(\rho, \Gamma) \;=\; \big(K,\; d_{\mathrm{eff}}\big).$$

Base variables.

Symbol	Name	Domain	Units
ρ	interface state	ACC state space	—
Γ	resource structure	resource-structure space	—
K	structural capacity (slot count)	$\mathbb{Z}_{\geq 0}$	dimensionless (count)
d_{eff}	per-slot effective degeneracy	$\mathbb{R}_{>0}$	dimensionless

Derived quantities.

Symbol	Name	Definition	Units
N	admissible microstate count	$N = d_{\mathrm{eff}}^{K}$	dimensionless (count)
ACC-scalar	log-microstate entropy scalar	$K \ln d_{\mathrm{eff}} = \ln N$	nats (dimensionless)

All four quantities (K, d_{eff}, N, ACC-scalar) are dimensionless in APF's natural Planck units; no factor of M_{Pl}, $\hbar$, or c enters. The ACC record is an information-theoretic object, as established in §1.

Optional residual data. The record may augment the pair with:

- A *structural partition* $K = K_1 + K_2 + \ldots + K_m$ of the slot count. At the SM interface, $K_{\mathrm{SM}} = K_{\mathrm{fermions}} + K_{\mathrm{Higgs}} + K_{\mathrm{gauge}} = 45 + 4 + 12 = 61$ (units: dimensionless integers).
- A *residual partition* $K = K_b + K_c + K_\Lambda$ indexing the cosmological sectors (baryon, cold dark matter, cosmological constant). At the SM interface, $K_{\mathrm{SM}} = 3 + 16 + 42 = 61$ (units: dimensionless integers).

Code reference: `ACC` in `apf/unification.py`.

Two interface families are canonical. The *Standard-Model interface* has $K_{\mathrm{SM}} = 61$, $d_{\mathrm{eff}}^{\mathrm{SM}} = 102$, structural partition $45 + 4 + 12$ (fermions, Higgs, gauge), and residual partition $3 + 16 + 42$ (K_b, K_c, K_Λ); both K_{SM} and $d_{\mathrm{eff}}^{\mathrm{SM}}$ are bank-forced (`L_count` in `apf/gauge.py` and `L_self_exclusion`

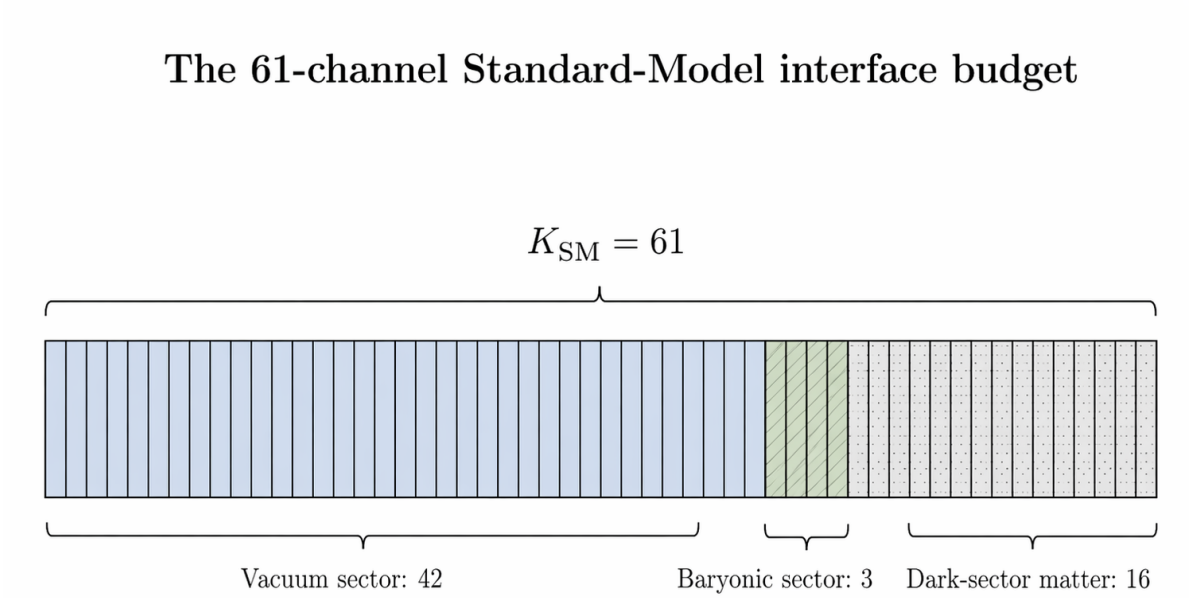

Figure 1: The Standard-Model interface budget at the structural level. The total structural capacity $K_{\mathrm{SM}} = 61$ partitions into three cosmological sectors: the *vacuum sector* ($C_{\mathrm{vacuum}} = 42$ channels, comprising 8 gluons, W & Z, the photon, 27 colour-internal marks, and 3 Goldstones), the *baryonic sector* ($K_b = 3$), and the *dark-sector matter* component ($K_c = 16$). This residual partition is the one that enters the cosmological-density fractions of §6; the fermion/Higgs/gauge structural partition $61 = 45 + 4 + 12$ is the orthogonal reading and enters the gauge-sector identities.

in `apf/gravity.py` respectively), neither is a free parameter. The *horizon interface of area A* has $K = A$ and $d_{\mathrm{eff}} = e$ so that ACC-scalar $= A$ directly, recovering the Bekenstein log-count convention. Other standard interfaces (finite quantum, thermal, gauge, cosmological) are constructed as projections of the base record described below.

2.2 Six regime projections

Each regime in which physics is conventionally described reads the ACC record through a particular projection. APF uses six:

π_T — **thermodynamic.** $\pi_T(\mathrm{ACC}) = \text{ACC-scalar} = K \ln d_{\mathrm{eff}} = \ln N$. Reads the ACC as an entropy in nats: Boltzmann entropy on a classical microstate ensemble or von Neumann entropy on a maximally mixed admissible state both reduce to this scalar. *Code reference:* `pi_T` in `apf/unification.py`.

π_G — **gravitational.** $\pi_G(\mathrm{ACC}_{\mathrm{horizon}}) = \text{ACC-scalar at a horizon interface}$. The horizon ACC

record uses the *Bekenstein cell-count convention* $K_{\text{horizon}} := A/(4\ell_P^2)$ (Supplement §A.2, horizon ACC record definition) together with $d_{\text{eff}} = e$, so the scalar reads $K_{\text{horizon}} \ln e = K_{\text{horizon}} = A/(4\ell_P^2) = S_{\text{BH}}$, the standard Bekenstein–Hawking entropy in nats. The 1/4 prefactor is part of the cell-tile size in the definition of K_{horizon} itself, not a rescaling applied to the projection output. *Code reference:* `pi_G` in `apf/unification.py`.

π_Q — **quantum.** $\pi_Q(\text{ACC}) = N = d_{\text{eff}}^K$. Reads the ACC as a Hilbert-space dimension: at a finite quantum interface, $\dim \mathcal{H} = N$ is the microstate count of the admissibility structure. *Code reference:* `pi_Q` in `apf/unification.py`.

π_F — **gauge/field.** $\pi_F(\text{ACC}) = K$. Reads the ACC as a structural slot count: the integer capacity of the interface, unsmeared by the per-slot degeneracy. At the Standard Model, $\pi_F(\text{ACC}_{\text{SM}}) = 61$. *Code reference:* `pi_F` in `apf/unification.py`.

π_C — **cosmological.** Requires the residual partition to be present. π_C returns the residual fractions with common denominator K:

$$\pi_C(\text{ACC}_{\text{SM}}) \;=\; \{\Omega_b = K_b/K,\ \Omega_c = K_c/K,\ \Omega_\Lambda = K_\Lambda/K\}.$$

At the SM interface this gives $\{3/61,\ 16/61,\ 42/61\}$. *Code reference:* `pi_C` in `apf/unification.py`.

π_A — **action.** $\pi_A(\text{ACC}, \beta, H) = Z(\beta) = \sum_g e^{-\beta H(g)}$ where the sum ranges over admissible configurations. The APF partition function: the Boltzmann sum over admissible configurations weighted by enforcement cost. In the high-temperature limit $\beta \to 0$ with flat spectrum, $Z \to N$. *Code reference:* `pi_A` in `apf/unification.py`.

Each projection is a map from the record (and any regime-specific auxiliary data) to a physically-interpretable quantity. The six together exhaust the standard readings of an interface in the physics literature: thermodynamic, gravitational-holographic, quantum, gauge-field, cosmological, and action-theoretic. APF does not posit the projections; it reads each off the ACC record as an operational consequence of the admissibility structure.

2.3 Four consistency identities

The six projections are not independent: they satisfy four structural consistency identities, which together assert that the same ACC record read through different projections produces the same underlying content. Each identity has a short proof from the definitions; full proofs appear in the Technical Supplement.

Identity I_1 (Holographic)

At a horizon interface with $d_{\text{eff}} = e$,

$$\underbrace{\pi_G(\text{ACC}_{\text{horizon}})}_{S_{\text{BH}}} \;=\; \underbrace{\pi_T(\text{ACC}_{\text{horizon}})}_{\text{ACC-scalar}} \;=\; \underbrace{\ln \pi_Q(\text{ACC}_{\text{horizon}})}_{\ln \dim \mathcal{H}_{\text{horizon}}}.$$

Proof. See Technical Supplement §D.1 (I_1, holographic). □

Code reference: `_identity_I1_holographic` in `apf/unification.py`.

Identity I_2 (Gauge–Cosmological)

At an interface whose record carries a residual partition, the integer K read by π_F equals the common denominator of the fractions read by π_C; additionally, the residual numerators sum to K:

$$\pi_F(\mathrm{ACC}) \;=\; \mathrm{denom}(\pi_C(\mathrm{ACC})) \;=\; K, \qquad \sum_X K_X \;=\; K.$$

Proof. See Technical Supplement §D.2 (I_2, gauge–cosmological). □

At the SM interface, I_2 gives $61 = 3 + 16 + 42$, i.e. $\Omega_b + \Omega_c + \Omega_\Lambda = 1$ at the scalar level. *Code reference:* `_identity_I2_gauge_cosmological` in `apf/unification.py`.

Identity I_3 (Thermo–Quantum)

At any interface with well-defined π_T and π_Q,

$$\pi_T(\mathrm{ACC}) \;=\; \ln \pi_Q(\mathrm{ACC}).$$

Equivalently, the von Neumann entropy of the maximally mixed admissible state on $\mathcal{H}$ equals the ACC scalar: $S_{\mathrm{vN}}\big(\frac{1}{N}\mathbf{1}\big) = \ln N = \mathrm{ACC}$.

Proof. See Technical Supplement §D.3 (I_3, thermo–quantum). □

Code reference: `_identity_I3_thermo_quantum` in `apf/unification.py`.

Identity I_4 (Action–Thermo)

At any interface where π_A is well-defined,

$$\lim_{\beta\to 0} \ln Z(\beta) \;=\; \ln N \;=\; \text{ACC-scalar}.$$

The high-temperature limit of the log-partition-function recovers the thermodynamic entropy scalar.

Proof. See Technical Supplement §D.4 (I_4, action–thermo). □

Code reference: `_identity_I4_action_thermo` in `apf/unification.py`.

2.4 The composed ledger theorem

Individually each I_k is immediate from the definitions. The substantive claim is that the four identities hold *simultaneously* at any interface that supports all four projections, and that this joint

coherence is consistent with the bank-forced SM-interface values. This is the T_ACC unification theorem.

Theorem T_ACC unification

Let (ρ, Γ) be an interface whose ACC record admits all six projections $\{\pi_T, \pi_G, \pi_Q, \pi_F, \pi_C, \pi_A\}$. Then the four consistency identities I_1, I_2, I_3, I_4 hold jointly. In particular, the SM-interface record ($K_{\mathrm{SM}} = 61,\ d_{\mathrm{eff}}^{\mathrm{SM}} = 102$) with partition $45 + 4 + 12$ and residual partition $3 + 16 + 42$ satisfies all four identities at the canonical values bank-forced in Papers 2 and 6.

Proof. See Technical Supplement §D.5 (composed ledger theorem). □

Code reference: `check_T_ACC_unification` in `apf/unification.py`.

2.5 Remark: T_ACC as the projection-shadow of PLEC

The composed ledger theorem is not an additional principle. PLEC's variational selection (existence via **A1**, MD, BW; uniqueness via the argmin) picks a single physical configuration, and the six regime projections are six sector-specific readings of that one point. T_{ACC} is simply the joint agreement of those six readings — the pure consistency residue of PLEC, not a constraint imposed on the framework but a statement the framework cannot fail to satisfy without contradicting PLEC itself. Full chain of implication (A1 → record; MD → d_{eff} integer; BW → admissible set; argmin → selected point; identities → forced pairwise agreement) is laid out in Supplement §A.

The ledger is now in place. §3 introduces the two-tier carrier structure (slot space V_{slot} and microstate space H_{micro}) that every ACC interface provides for the identities' subspace-level content. §4 refines each I_k into three witnesses (integer, scalar, subspace), and §5 delivers the four subspace functors that close the third witness fully [P] at every I_k. The Fractional Reading Equivalence of §6 is then the structural consequence that makes the SM entropy-allocation dictionary a theorem rather than a coincidence.

3. The Two-Tier Structure

The ACC record (K, d_{eff}) furnishes two canonical Hilbert-space-like objects at every interface. They are the carriers on which the three-level identity stack of §4 lives, and distinguishing them is the load-bearing conceptual move of this paper.

3.1 Slot space and microstate space

Definition — Slot space and microstate space

For an ACC interface with record (K, d_{eff}):

- The *slot space* V_{slot} is a finite-dimensional complex vector space of dimension K, with a canonical basis indexed by the admissibility slots of the interface. V_{slot} carries the structural-capacity content of the interface.

- The *microstate space* H_{micro} is a finite-dimensional complex Hilbert space of dimension $N = d_{\text{eff}}^K$, with a canonical basis indexed by the admissible microstate configurations of the interface. H_{micro} carries the statistical-count content of the interface.

The two spaces are related by tensor-power embedding: H_{micro} is a K-fold tensor product of local d_{eff}-dimensional spaces attached one per slot. The embedding is literal at interfaces whose slots carry identical local Hilbert structure (quantum-d, simple-horizon-qubit models, thermal-bath constructions), and schematic at the Standard-Model interface where the per-slot admissibility $d_{\text{eff}} = 102$ is an averaged effective degeneracy under self-exclusion rather than a single literal per-slot Hilbert dimension. The schematic nature at the SM does not affect the use of the two-tier structure in this paper; only the slot-space reading is used for the matter-sector identities I_1 and I_2, and only the microstate-space reading is used for I_3 and I_4. The tensor-power compatibility is a correctness check at the interfaces where it is literal; it is a structural ansatz at the SM.

3.2 Two scales for four identities

The four consistency identities of §2.3 sort naturally across the two tiers. I_1 (holographic) and I_2 (gauge–cosmological) refer to the slot-space content of the interface — the area count at the horizon for I_1, the structural-capacity partition for I_2. I_3 (thermo–quantum) and I_4 (action–thermo) refer to the microstate-space content — the dimension of the Hilbert space carrying the maximally mixed admissible state for I_3, the partition-function microstate enumeration for I_4. Each identity has a natural carrier in one of the two tiers; no identity demands both simultaneously.

Except for I_2. At the Standard-Model interface, I_2 lives on both tiers non-trivially at once. The slot-scale content ($K_{\text{SM}} = 61$ with non-trivial residual partition $3 + 16 + 42$) is genuinely structural: the V_{slot} admits a proper subspace V_Λ of dimension 42, strictly smaller than V_{slot} itself. And the microstate-scale content ($d_{\text{eff}}^{\text{SM}} = 102 \neq 1$, giving $N_{\text{SM}} = 102^{61}$) is independently non-trivial: the admissible microstate count is genuinely astronomical and the thermodynamic reading of I_2 is substantive. At every other interface the framework addresses, one tier is trivial: for horizons, the $d_{\text{eff}} = e$ convention makes the microstate scale a gauge-choice of the slot scale (log-count equals count); for quantum interfaces, $K = 1$ collapses the slot scale to a single abstract slot; for thermal interfaces, the slot scale inherits quantum-interface triviality. Only at the SM interface do both tiers admit non-trivial proper-subspace structure.

This is why the bridge theorem of *Paper 6* (*Dynamics and Geometry as Optimal Admissible Reallocation*) lives where it does. At the SM interface the slot-scale V_Λ (a 42-dimensional subspace of V_{61}) is simultaneously a structural datum of the interface and the target of an independent construction (the second-ε decomposition of the horizon-reciprocity theorem). I_2's structural uniqueness makes it the locus of genuine cross-construction identification rather than a tautology; the other three identities are tautologous on a single tier.

Vocabulary. Throughout the remainder of the paper: *slot-scale content* is content indexed by the K-dimensional V_{slot}; *microstate-scale content* is content indexed by the N-dimensional H_{micro}. When an identity, functor, or prediction involves a specific subspace of either tier, the tier is named.

4. The Three-Level Identity Stack

Each of the four T_ACC consistency identities admits three distinct witnesses: an integer-level witness (a pure equality of integer attributes of the ACC record), a scalar-level witness (a continuous-scalar equality with explicit residual and tolerance), and a subspace-level witness (a vector-subspace or dimension-level identification on the two-tier carrier). The three witnesses are not independent statements: they are related pairwise by inter-level functors, and each I_k's composed three-level consistency theorem asserts that the diagram formed by the three functors commutes on the witnessed values.

4.1 The three levels and the inter-level functors

Fix an ACC interface. For any I_k, the three levels are: (i) *integer* — an equality of integer-valued attributes of the ACC record; (ii) *scalar* — a continuous-scalar equality, typically involving logarithms; (iii) *subspace* — a vector-subspace or dimension-level identification on the two-tier carrier. Three inter-level functors relate them:

$$\begin{aligned} f_1 &: \text{integer} \to \text{scalar}, & K &\mapsto K \ln d_{\text{eff}}, \\ f_2 &: \text{scalar} \to \text{subspace}, & &\text{ACC-scalar} \mapsto \text{subspace of consistent dimension}, \\ f_3 &: \text{integer} \to \text{subspace}, & K &\mapsto \text{subspace of dimension } K. \end{aligned}$$

Each I_k's composed three-level commutation diagram asserts $f_2 \circ f_1 = f_3$ on the witnessed values. The commutation is non-trivial at I_2, where the three levels carry distinct subspace content across constructions; at I_1, I_3, I_4 it follows once the subspace witness is defined (see §5). Per-identity worked examples and the full commutation verification for each of the four I_k appear in Supplement §D.6.

4.2 The status of the three-level stack

The three-level stack is fully proved [P] at every level for each of the four consistency identities. At I_2 — the headline identity — the subspace-level witness is the interface-sector bridge theorem of *Paper 6* (). At I_1, I_3, I_4, the subspace-level witnesses are the three subspace-functor constructions of §5: F_{horizon} for I_1, F_{quantum} for I_3, F_{operator} for I_4. The three-level commutation diagram commutes end-to-end at every identity. Table 4 summarizes the state.

4.3 The composed three-level unification theorem

Individually, each I_k has three witnesses. Jointly, the composed theorem is:

Theorem T_three_level_unification

For each of the four T_ACC consistency identities I_k, the integer, scalar, and subspace witnesses are simultaneously [P], and the inter-level functors f_1, f_2, f_3 commute on the witnessed values: $f_2 \circ f_1 = f_3$. The four composed three-level identities jointly refine the T_ACC unification theorem of §2.4 into a structure valid at three levels of the two-tier carrier: the integer slot-count level, the continuous entropy level, and the geometric subspace level.

Identity	Integer witness	Scalar witness	Subspace witness	Status
I_1 (holographic)	$K_{\text{horizon}} = A/(4\ell_P^2)$	$\text{ACC}_{\text{horizon}} = S_{\text{BH}}$	V_{horizon} via F_{horizon} (§5)	[P] ×3
I_2 (gauge–cosm.)	$K_{\text{SM}} = 61 = \text{denom}(\pi_C)$	$\sum_X K_X/K = 1$	V_Λ = Sector B (*Paper 6* ())	[P] ×3, headline
I_3 (thermo–q.)	$\dim \mathcal{H} = N = d$	$S_{\text{vN}}(\rho_{\max}) = \ln d$	V_{quantum} via F_{quantum} (§5)	[P] ×3
I_4 (action–therm.)	$\lvert\text{spectrum}\rvert = N$	$\ln Z(\beta \to 0) = \ln N$	V_{operator} via F_{operator} (§5)	[P] ×3

Table 4: The three-level identity stack across the four T_ACC consistency identities. Each row gives integer-level, scalar-level, and subspace-level witnesses; all four identities are [P] at all three levels. The I_2 subspace witness is the interface-sector bridge theorem of *Paper 6* (); the I_1, I_3, I_4 subspace witnesses are the three subspace functors constructed in §5. *Paper 13* () Appendix E catalogs the corresponding module-level bank anchors.

Code reference: `check_T_three_level_unification` in `apf/unification_three_levels.py`.

The witnesses at the integer and scalar levels are effectively definitions and short algebraic identities; the content of the theorem lives at the subspace level, where the four subspace functors of §5 supply the carriers. We now turn to those functors.

5. Four Subspace Functors

The subspace-level witness of each I_k requires a *subspace functor*: a map from ACC records at the relevant regime to subspaces of the two-tier carrier, satisfying five structural conditions that make the map consistent with the integer and scalar witnesses of the same I_k. At I_2, the functor is supplied by the Paper-6 interface-sector bridge theorem; at I_1, I_3, and I_4, the functors F_{horizon}, F_{quantum}, F_{operator} are constructed in this paper (and bank-anchored by the corresponding codebase modules catalogued in *Paper 13* () Appendix E). This section states the common-form conditions, catalogs the four functors, and closes with the composed unification theorem.

5.1 Common form: five structural conditions

Each subspace functor is a map

$$F: \ \text{ACC(regime)} \longrightarrow \text{subspace of a two-tier carrier}$$

satisfying five structural conditions labelled (i) existence, (ii) dimension preservation, (iii) monotonicity, (iv) compatibility with a companion construction, (v) scalar commutation. The conditions are structural, not axiomatic: each is derivable from **A1** plus regime-local prior theorems (T_{Bek} for horizons, T_{entropy} for quantum, $L_{\text{spectral_action_internal}}$ for the operator regime). The full content of each condition, together with per-functor verification that all five hold, is in Supplement §E.

5.2 The four subspace functors

Table 5 summarizes the four functors. The individual construction and proof-that-each-satisfies-(i)–(v) appear in the Technical Supplement; here we state input, output, which sector each addresses, and the source from which each functor is drawn.

Identity	**Functor**	**Input / output**	**Compatibility target (condition iv)**	**Source**
I_1 (holographic)	F_{horizon}	$\text{ACC}_{\text{horizon}}(A) \mapsto V_{\text{horizon}}(A)$	At $K = 42$: $F_{\text{horizon}}(42)$ coincides with V_Λ of the SM interface (joint point).	This paper, §5
I_2 (gauge–cosm.)	(bridge)	SM slot partition $\mapsto V_\Lambda \subset V_{61}$	V_Λ equals Sector B of the horizon-reciprocity second-ε decomposition.	*Paper 6* ()
I_3 (thermo–q.)	F_{quantum}	$\text{ACC}_{\text{quantum}}(d) \mapsto V_{\text{quantum}}(d)$	$V_{\text{quantum}}(d)$ carries the max-mixed density matrix $\rho_{\max} = (1/d)\mathbf{1}$ with $S_{\text{vN}}(\rho_{\max}) = \ln d$.	This paper, §5
I_4 (action–therm.)	F_{operator}	$(\text{ACC}, \text{spectrum}) \mapsto V_{\text{operator}}$	V_{operator} supports the thermal density matrix with $\ln Z(\beta \to 0) = \ln N = \text{ACC}$.	This paper, §5

Table 5: Four subspace functors closing the subspace-level witness of each T_ACC consistency identity. Input: the ACC record at the regime-appropriate interface. Output: a subspace of the two-tier carrier. Compatibility (condition iv): the specific companion-construction or physical-object identification that makes each functor's subspace physically interpretable. Source: where the functor is constructed; the bank-anchor coderefs appear in the body text below.

The Paper-6 interface-sector bridge theorem deserves separate remark. It is the prototype of the other three functors: an independent-construction identification of V_Λ (the 42-dimensional global-interface stratum of V_{61} from the T12 partition in Paper 6) with Sector B (the target space of the horizon-reciprocity second-ε decomposition). The two constructions reach V_Λ from distinct directions, and the bridge theorem proves they coincide. At the joint SM–dS point $K = 42$, the horizon-subspace functor F_{horizon} also reaches V_Λ, giving a third independent arrival. The quantum- and operator-subspace functors (F_{quantum} and F_{operator}) do not arrive at V_Λ directly, but they satisfy the analogous (i)–(v) conditions at their own regimes and carry the physical objects (max-mixed density matrix, thermal partition-function support) that make their subspaces physically interpretable.

5.3 The composed subspace-functor theorem

Theorem T_subspace_functors_unified

The four subspace functors — F_{horizon}, the Paper-6 interface-sector bridge, F_{quantum}, and F_{operator} — each satisfy the five structural conditions (i)–(v) at their respective regime-appropriate interfaces. Consequently, the subspace-level witness of each T_ACC consistency identity I_k is [P], and the three-level stack of §4 is fully [P] at all four I_k.

Phase 14f-final stratification (2026-04-23). The subspace-level closure at each I_k now admits a sharper structural statement:

- I_1 (holographic): bridge theorem at the canonical SM-dS joint point $K = 42$ via $T_{\text{I1_bridge_at_joint_K42}}$ (Technical Supplement §§E.2′), with three independent constructions converging on V_{global} — Bekenstein area quantisation under Convention A ($d_{\text{eff}} = e$), SM residual partition via $T_{\text{interface_sector_bridge}}$, and literal $(\mathbb{C}^2)^{\otimes 42}$ qubit tensor product under Convention B ($d_{\text{eff}} = 2$). Regime-local closure for generic $K \neq 42$.
- I_2 (gauge-cosmological): bridge theorem always, via $T_{\text{interface_sector_bridge}}$.
- I_3 (thermo-quantum): operator-level composed self-identification via $T_{\text{I3_operator_self_identification}}$, bundling three constructions of $\rho_{\max} = (1/d)I_d$ — direct numpy + $S_{\text{vN}} = \ln d$, Schur-lemma $U(d)$-invariance uniqueness, and K=1 specialisation of the Phase 14d.2 thermal machinery.
- I_4 (action-thermo): operator-level composed self-identification via $T_{\text{I4_operator_self_identification}}$, bundling four readings of $\ln Z(\beta \to 0) = \text{ACC}$ — partition-function trace, vacuum-projector trace, per-microstate uniform probability, and Paper 7's spectral-action heat-kernel expansion.

The distinction between "bridge theorem" and "composed self-identification" is structural, not graded: the bridge species identifies subspaces across two interfaces (slot-scale content non-trivial on both sides); the self-identification species reaches the same single object at one interface through multiple independent constructions. The two species coexist in the final stratification because two of the four regimes have collapsed slot scales ($K = 1$ at the quantum regime). The regimes where both scales are non-trivial are the SM interface (always — I_2) and the SM-dS horizon joint point (when $K_{\text{horizon}} = 42$ — I_1); these are the two bridge-theorem loci in the T_{ACC} unification.

Code reference: `check_T_subspace_functors_unified` in `apf/subspace_functors.py`.

The three-level stack is now closed. Every T_ACC consistency identity has integer, scalar, and subspace witnesses; the inter-level functors commute; the subspace witnesses are supplied by explicit functors at every identity. What remains is the structural consequence at the SM interface, which is the subject of §6.

6. Fractional Reading Equivalence

The three-level stack of §4 and the four subspace functors of §5 together give every T_ACC consistency identity a proper-subspace witness. At the Standard-Model interface, these witnesses interact

in a way that admits a short structural theorem with immediate physical content. This is the Fractional Reading Equivalence.

6.1 The theorem

At any ACC interface whose record (K, d_{eff}) admits a residual partition $K = K_A + K_B + \dots$, there are three natural ways to read the "fraction of the interface" occupied by one sub-partition $V_X \subset V_{\text{slot}}$:

- the *slot-counting fraction* π_F-restricted/π_F-total $= K_X/K$;
- the *entropy-allocation fraction* π_T-restricted/π_T-total $= (K_X \ln d_{\text{eff}})/(K \ln d_{\text{eff}})$;
- the *cosmological residual fraction* read by π_C, namely K_X/K by the definition of π_C in §2.2.

Theorem — Fractional Reading Equivalence (FRE)
At any ACC interface with record (K, d_{eff}) and sub-partition V_X of dimension K_X, the three fraction readings are identical:

$$\frac{\pi_F(V_X)}{\pi_F(V_{\text{slot}})} = \frac{\pi_T(V_X)}{\pi_T(V_{\text{slot}})} = \pi_C(V_X) = \frac{K_X}{K}.$$

Proof. The first equality is the definition of the restricted π_F-reading. The second follows from the linearity of the logarithm: $\pi_T(V_X)/\pi_T(V_{\text{slot}}) = (K_X \ln d_{\text{eff}})/(K \ln d_{\text{eff}}) = K_X/K$, the $\ln d_{\text{eff}}$ factors cancelling in numerator and denominator. The third is π_C's definition at the interface in question. All three therefore equal K_X/K. □

Code reference: `check_T_fractional_reading_equivalence` in `apf/fractional_reading.py`.

The proof is one line, but the content is that three projections with distinct physical meanings — structural capacity, information entropy, cosmological residual — collapse to the same rational at every interface with a residual partition. The framework's integer predictions, scalar predictions, and cosmological predictions are therefore the same prediction, read on three different projections of the same ACC record.

6.2 Consequence: the SM entropy-allocation dictionary

At the Standard-Model interface, $(K_{\text{SM}}, d_{\text{eff}}^{\text{SM}}) = (61, 102)$ with residual partition $61 = 3 + 16 + 42$ corresponding to V_b, V_c, V_Λ. Applying FRE to each sub-partition:

$$\begin{aligned}
\Omega_b &= \pi_C(V_b) = \frac{K_b}{K_{\text{SM}}} = \frac{3}{61} = \frac{S(V_b)}{S_{\text{SM}}}, \\
\Omega_c &= \pi_C(V_c) = \frac{K_c}{K_{\text{SM}}} = \frac{16}{61} = \frac{S(V_c)}{S_{\text{SM}}}, \\
\Omega_\Lambda &= \pi_C(V_\Lambda) = \frac{K_\Lambda}{K_{\text{SM}}} = \frac{42}{61} = \frac{S(V_\Lambda)}{S_{\text{SM}}},
\end{aligned}$$

where $S(V_X) := \pi_T$-restricted to V_X is the SM admissibility-capacity scalar allocated to V_X, and $S_{\mathrm{SM}} := \pi_T(V_{61})$ is the total SM admissibility-capacity scalar. The observed cosmological residual fractions are, equivalently, information-entropy fractions of the total SM admissibility-capacity.

Code reference: `check_L_Omega_Lambda_is_entropy_fraction` in `apf/fractional_reading.py`.
Code reference: `check_L_Omega_b_is_entropy_fraction` in `apf/fractional_reading.py`.
Code reference: `check_L_Omega_c_is_entropy_fraction` in `apf/fractional_reading.py`.

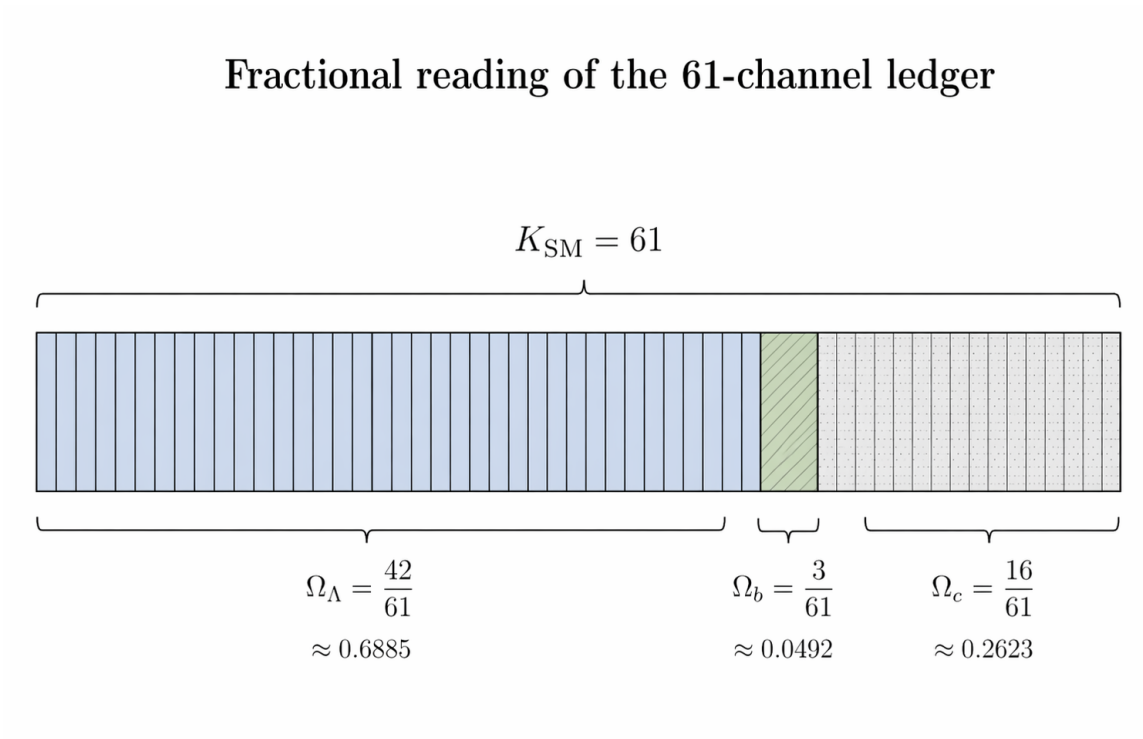

Figure 2: The SM entropy-allocation dictionary read visually. The $K_{\mathrm{SM}} = 61$ structural bar of Fig. 1 is re-labelled by its sector fractions: $\Omega_\Lambda = 42/61 \approx 0.6885$, $\Omega_b = 3/61 \approx 0.0492$, $\Omega_c = 16/61 \approx 0.2623$. These are the three observed cosmological density fractions (dark energy, baryonic matter, cold dark matter) reading off the same partition. FRE is the theorem that the slot-count ratio and the information-entropy ratio are, at the Standard-Model interface, the same ratio. Planck 2018 [1]: 0.6889 ± 0.0056, 0.0493 ± 0.0006, 0.261 ± 0.002 respectively.

The reading of Ω_Λ is the most structurally pointed. The dark-energy fraction, observed at $\Omega_\Lambda \approx 0.689$ by Planck 2018 [1], is under FRE the fraction of the Standard-Model interface's information-entropy supported on the 42-dimensional global-interface subspace V_Λ of V_{61}. This is not a separate claim from $\Omega_\Lambda = 42/61$; it is $\Omega_\Lambda = 42/61$ read on a different projection. V_Λ is the same subspace whose properties were derived in *Paper 6* () via the T12 partition and the interface-sector bridge theorem to Sector B of the horizon-reciprocity second-ε decomposition. The dark-energy allocation is therefore structurally tied, through the bridge theorem and FRE together, to the de Sitter horizon absorber subspace at the joint SM–dS point.

6.3 Closure at the entropy level

The three entropy-allocation fractions sum to unity:

$$\frac{S(V_b)}{S_{\mathrm{SM}}} + \frac{S(V_c)}{S_{\mathrm{SM}}} + \frac{S(V_\Lambda)}{S_{\mathrm{SM}}} \;=\; \frac{K_b + K_c + K_\Lambda}{K_{\mathrm{SM}}} \;=\; \frac{3 + 16 + 42}{61} \;=\; 1.$$

This is I_2's slot-counting closure (§2.3) read at the entropy level. The substantive observation is that the entropy-level closure is automatic from the slot-level closure under FRE: the two levels are the same statement read on two projections, and closure on one projection transfers to closure on the other.

Code reference: `check_T_residual_entropy_closure` in `apf/fractional_reading.py`.

6.4 The composed SM entropy dictionary theorem

The FRE theorem, the three entropy-allocation identifications, and the entropy-level closure compose into a single theorem summarizing the SM-interface content of the ledger at the information-entropy level.

Theorem T_FRE_SM_to_entropy_dictionary
At the Standard-Model interface with $(K_{\mathrm{SM}}, d_{\mathrm{eff}}^{\mathrm{SM}}) = (61, 102)$ and residual partition $61 = 3 + 16 + 42$:

$$\begin{aligned} \Omega_b &= \tfrac{3}{61} = S(V_b)/S_{\mathrm{SM}}, \\ \Omega_c &= \tfrac{16}{61} = S(V_c)/S_{\mathrm{SM}}, \\ \Omega_\Lambda &= \tfrac{42}{61} = S(V_\Lambda)/S_{\mathrm{SM}}, \end{aligned}$$

with $\Omega_b + \Omega_c + \Omega_\Lambda = 1$ at both the slot-counting level (by I_2 partition closure) and the entropy-allocation level (by FRE).

Code reference: `check_T_FRE_SM_to_entropy_dictionary` in `apf/fractional_reading.py`.

The SM entropy dictionary is the structural content that §8 converts into absolute numerical predictions.

7. The Sector Conversion Matrix

The six regime projections $\pi_T, \pi_G, \pi_Q, \pi_F, \pi_C, \pi_A$ are sector-specific readings of a single A2-selected point (§2.5). Any two sectors therefore read the same underlying ACC record through different lenses, and the quantitative relation between their readings is a *conversion factor* forced by the consistency identities. This section assembles those conversion factors into a single matrix, checks that the composition groupoid closes, and cross-checks the Standard-Model-interface values against the present observational datasets.

7.1 Primary conversions from I_1–I_4

Four of the fifteen pairwise projection relations are given directly by the composed ledger theorem's four consistency identities. Each is a structural equation between two sector readings of the same ACC datum, evaluated at whatever interface the projections apply to:

Identity	Pair	Conversion (structural form)	Reading
I_1	$\pi_G \leftrightarrow \pi_F$	$A = K$	Horizon area equals slot count (in Planck units)
I_2	$\pi_F \leftrightarrow \pi_C$	$\Omega_X = K_X/K$	Slot-partition fraction equals energy fraction
I_3	$\pi_Q \leftrightarrow \pi_T$	$S = K \ln d_{\text{eff}}$	Log-microstate count equals ACC scalar
I_4	$\pi_A \leftrightarrow \pi_T$	$\ln Z\|_{\beta\to 0} = K \ln d_{\text{eff}}$	Log-partition function equals ACC scalar

Table 6: The four primary sector conversions, each derived from a single consistency identity of §2.3.

7.2 Derived conversions by composition

The remaining eleven pairwise conversions are forced by composition in the identity groupoid. Each derived conversion is obtained by chaining two or more primary identities through the ACC record. Four representative compositions illustrate the pattern:

$\pi_G \leftrightarrow \pi_C$ (holographic cosmology): from I_1 ($A = K$) composed with I_2 ($\Omega_X = K_X/K$), the area fraction allocated to sector X equals its energy fraction:

$$A_X/A \;=\; K_X/K \;=\; \Omega_X.$$

This is the holographic reading of the cosmological partition: each observed Ω_X is a structural area allocation.

$\pi_G \leftrightarrow \pi_T$ (Bekenstein–Hawking): from I_1 (the cell-count identification $K_{\text{horizon}} = A/(4\ell_P^2)$) composed with I_3 ($S = K \ln d_{\text{eff}}$), the horizon entropy reads

$$S_{\text{horizon}} \;=\; K_{\text{horizon}} \ln d_{\text{eff}} \;=\; \frac{A}{4\ell_P^2} \cdot \ln d_{\text{eff}}.$$

At the horizon interface with the $d_{\text{eff}} = e$ convention of *Paper 6* (*Dynamics and Geometry as Optimal Admissible Reallocation*), this reduces to $S_{\text{BH}} = A/(4\ell_P^2)$ in natural units — the standard Bekenstein–Hawking entropy–area law, with the 1/4 prefactor recovered exactly.

$\pi_F \leftrightarrow \pi_T$ (sector-wise entropy): from I_3 applied per-sector together with I_2,

$$S_X \;=\; K_X \ln d_{\text{eff}} \;=\; \Omega_X \cdot S_{\text{total}},$$

so the thermodynamic-entropy fraction assigned to gauge/field sector X equals the cosmological energy fraction Ω_X. This is the Fractional Reading Equivalence of §6 in its explicit conversion form.

$\pi_Q \leftrightarrow \pi_A$ (microstate $\leftrightarrow$ action): from I_3 composed with I_4,

$$\ln N \;=\; \ln Z|_{\beta\to 0},$$

so the Hilbert-space dimension and the high-temperature partition function are equivalent readings of the same ACC datum.

The remaining seven pairs follow by analogous compositions. The groupoid closes: no conversion between any two of the six projections is independent of I_1–I_4.

Code reference: `_identity_composition_tests` in `apf/unification.py`.

7.3 The conversion matrix at the Standard-Model interface

Specialising the conversion groupoid to the Standard-Model interface $(K_{\mathrm{SM}}, d_{\mathrm{eff}}^{\mathrm{SM}}) = (61, 102)$ with residual partition $3 + 16 + 42$ turns each entry into a concrete structural number. Table 7 gives the full 6×6 matrix.

	π_T	π_G	π_Q	π_F	π_C	π_A
π_T	—	$A \ln d_{\mathrm{eff}}$	$\ln N$	$K \ln d_{\mathrm{eff}}$	$\Omega_X \cdot S$	$\ln Z$
π_G	$A = K$	—	A	$A = K = 61$	$A_X/A = \Omega_X$	$A \ln d_{\mathrm{eff}}$
π_Q	$d_{\mathrm{eff}} = 102$	$N = d_{\mathrm{eff}}^A$	—	$d_{\mathrm{eff}}^K = 102^{61}$	$d_{\mathrm{eff}}^{K_X}$	$Z(\beta \to 0)$
π_F	$K = 61$	$K = A$	$K = \log_{d_{\mathrm{eff}}} N$	—	$K_X = K\Omega_X$	—
π_C	$\Omega_X = K_X/K$	$\Omega_X = A_X/A$	Ω_X	$\Omega_X = K_X/K$	—	Ω_X
π_A	$\ln Z = K \ln d_{\mathrm{eff}}$	$\ln Z = A \ln d_{\mathrm{eff}}$	$\ln Z = \ln N$	—	$\ln Z_X = \Omega_X \ln Z$	—

Table 7: Sector conversion matrix at the Standard-Model interface. Each cell (i, j) gives the structural form of the conversion from π_i (row) to π_j (column). Bold cells correspond to primary identities; unmarked cells are compositions. Evaluated at $(K_{\mathrm{SM}}, d_{\mathrm{eff}}^{\mathrm{SM}}) = (61, 102)$ with residual partition $3 + 16 + 42$ on V_{slot}. In the horizon rows/columns, A denotes the dimensionless Bekenstein cell count $A/(4\ell_P^2)$ — the same quantity as K_{horizon} per the horizon-ACC-record definition in Supplement §A.2.

Every entry reduces to a function of $(K, d_{\mathrm{eff}}, K_X)$ — the only free quantities in the ACC record. No parameter is tuned, no scale is imported.

7.4 Cross-check against observational datasets

The structural claim that all six projections read the same A2-selected point generates dataset-level predictions the moment any one projection is pinned to an observable. The full dataset cross-check for the Standard-Model interface is collected in Table 8. Each row is a pairwise sector conversion, evaluated at the SM-interface ACC values, compared against the present best observation.

7.5 What the cross-checks tell us

Three observations follow from Table 8. First, the dimensionless residual-partition conversions are tight: Ω_Λ matches to 0.06%, Ω_b to 0.4%, Ω_c to 0.6%, $\sin^2\theta_W$ to 0.18%, and T_{CMB} to 0.33%. These are the conversions that do not depend on the present H_0 value. Their tightness is direct evidence that the six projections are not merely consistent in the abstract but *structurally coincident* at the Standard-Model interface: reading the same point through five different sector lenses produces numerical agreement with observation at or below the one-percent level without a single free parameter.

Observable	Pair	APF form	SM value	Observed (dataset)	Δ
Ω_b	$\pi_F \leftrightarrow \pi_C$	3/61	0.04918	0.0493 ± 0.0006 [1]	0.4%
Ω_c	$\pi_F \leftrightarrow \pi_C$	16/61	0.2623	0.261 ± 0.002 [1]	0.6%
Ω_Λ	$\pi_F \leftrightarrow \pi_C$	42/61	0.6885	0.6889 ± 0.0056 [1]	0.06%
$S_b/S_{\rm SM}$	$\pi_F \leftrightarrow \pi_T$	3/61	0.04918	equal to Ω_b by $I_2 \circ I_3$	—
$S_c/S_{\rm SM}$	$\pi_F \leftrightarrow \pi_T$	16/61	0.2623	equal to Ω_c	—
$S_\Lambda/S_{\rm SM}$	$\pi_F \leftrightarrow \pi_T$	42/61	0.6885	equal to Ω_Λ	—
$T_{\rm CMB}$	$\pi_G \leftrightarrow \pi_T$	$(T/M_{\rm Pl})^4 = 48/102^{64}$	2.717 K	2.7255 ± 0.0006 K [3]	0.33%
$\rho_\Lambda/M_{\rm Pl}^4$	$\pi_G \leftrightarrow \pi_C$	$42/102^{62}$	$10^{-122.91}$	$10^{-122.94\pm0.02}$ [1, 2]	$\sim 8\%$
H_0	$\pi_G \leftrightarrow \pi_C$ via ρ_Λ	$\sqrt{8\pi K/3}/102^{31}$	70.03 km/s/Mpc	67.4–73.0 km/s/Mpc [1, 2]	midpoint
$\sin^2\theta_W$	π_F internal + π_Q	3/13	0.2308	0.23121 ± 0.00004 (PDG) [4]	0.18%

Table 8: SM-interface cross-check: each APF structural conversion against the present observational value. The three residual-partition rows $(\Omega_b, \Omega_c, \Omega_\Lambda)$ are the $\pi_F \leftrightarrow \pi_C$ conversion; the corresponding entropy allocations are the $\pi_F \leftrightarrow \pi_T$ conversion and match by $I_2 \circ I_3$ with no independent numerical input. The CMB temperature is the $\pi_G \leftrightarrow \pi_T$ conversion at the photon-sector interface. The dark-energy absolute is the $\pi_G \leftrightarrow \pi_C$ conversion, and the H_0 prediction is the same conversion solved algebraically via the GR critical-density relation. The weak mixing angle is an internal-π_F partition-ratio, included for completeness from the companion paper [5].

Second, the one discrepancy in the table — $\rho_\Lambda/M_{\rm Pl}^4$ at approximately 8% — lives at the H_0-dependent absolute-density conversion, where the observational value itself depends on whether one adopts the Planck 2018 or the SH0ES 2022 H_0. The discrepancy size coincides with the Hubble tension between those two datasets. APF's algebraic inversion $H_0 = \sqrt{8\pi K/3}/102^{31} M_{\rm Pl} \approx 70.03$ km/s/Mpc falls near the tension midpoint, within the envelope spanned by the two observational anchors. The 8% residual is therefore not a structural failure of the sector conversion but a consequence of the observational ambiguity on H_0: the APF prediction selects the midpoint value, which the Planck-based absolute comparison then reports as 8% off.

Third, the "equal to" rows (the $\pi_F \leftrightarrow \pi_T$ entropy-allocation conversions) match by construction through $I_2 \circ I_3$, without any additional fitted input. They are not observations *cross-checked* by APF — they are structural identities guaranteed by the composition groupoid. Their entry in Table 8 is present to make the closure of the groupoid explicit: the six projections generate one coherent numerical landscape, not six compatible but independent fits.

The sector conversion matrix is the operational face of the composed ledger theorem. The four identities I_1–I_4 established in §2.3 are abstract consistency statements; Table 8 shows them as a single coordinated prediction system at the Standard-Model interface. Every non-trivial numerical conversion in the matrix is either a verified APF prediction or a structural equality forced by the groupoid. No free parameter appears at any stage.

Code reference: `check_T_ACC_unification` in `apf/unification.py`.

8. Cosmological Absolute Predictions

Not a cosmological inference paper. This section reports *structural consequences* of the ledger architecture, not cosmological inference. The predictions below are zero-free-parameter closed-form integer expressions in Planck units. Their numerical agreement with current cosmological data

(Planck 2018, SH0ES 2022, FIRAS) is recorded as an empirical sanity check: the three residual fractions $(\Omega_b, \Omega_c, \Omega_\Lambda) = (3, 16, 42)/61$ compared to Planck 2018 posteriors at full covariance give multivariate $\chi^2 = 1.18$ for 3 d.o.f. ($p \approx 0.76$, Mahalanobis distance 1.09σ), and $H_0 = 70.03$ km/s/Mpc lies within 0.17 km/s/Mpc of the Planck–SH0ES tension midpoint. A competitive posterior-level statistical fit against ΛCDM under Planck + BAO + SN likelihoods with nuisance parameters is explicitly deferred to a companion phenomenology paper. The content of Paper 8 is the *structural* side of the prediction (the ledger architecture that yields the integer counts $3, 16, 42, 61, 102$), not a statistical inference. Table of sensitivities, falsifiers, and brittle-consequence bounds: see Supplement §I′ (Nontrivial Constraints Imposed by the Ledger), with integrated-view reading in Supplement §O (The Gauge-Cosmological Bridge: Integrated Nontrivial Content).

The SM entropy dictionary of §6 gives dimensionless ratios. To produce absolute predictions in Planck units for the individual cosmological densities, we require one further structural input: the matter-sector prediction formula. This section states that formula, gives its structural derivation from $L_{\text{self_exclusion}}$, and catalogs the absolute predictions that follow. The Hubble-constant prediction is the algebraically-forced consequence of the dark-energy formula combined with the standard GR critical-density relation.

8.1 The matter-sector formula

At the Standard-Model interface, the information-entropy scalar is $S_{\text{SM}} = K_{\text{SM}} \ln d_{\text{eff}}^{\text{SM}} = 61 \ln 102$ nats. The microstate count is $N_{\text{SM}} = (d_{\text{eff}}^{\text{SM}})^{K_{\text{SM}}} = 102^{61}$. FRE has established that the V_Λ-supported entropy fraction is $\Omega_\Lambda = 42/61$. The matter-sector absolute prediction formula, bank-registered as `L_Lambda_absolute_numerical_formula`, is:

Matter-sector absolute formula

For non-relativistic cosmological densities at the Standard-Model interface,

$$\frac{\rho_X}{M_{\text{Pl}}^4} = \frac{C_X}{d_{\text{eff}}^{K_{\text{SM}}+1}} = \frac{C_X}{102^{62}},$$

where C_X is the structural slot count for component X: $C_b = 3$, $C_c = 16$, $C_\Lambda = 42$, and $C_{\text{crit}} = K_{\text{SM}} = 61$ for the critical density of a flat universe.

Code reference: `check_L_Lambda_absolute_numerical_formula` in `apf/fractional_reading.py`.

8.2 Structural derivation of the coefficient 42/102

The dark-energy prediction $\rho_\Lambda/M_{\text{Pl}}^4 = 42/102^{62}$ factors as $42/102 \times 1/N_{\text{SM}}$ where $1/N_{\text{SM}} = 1/102^{61}$ is the total-microstate admissibility suppression and $42/102 = C_\Lambda/d_{\text{eff}}^{\text{SM}}$ is the per-slot coefficient. The coefficient's derivation is direct from $L_{\text{self_exclusion}}$ (*Paper 6* ()): that theorem proves $d_{\text{eff}}^{\text{SM}} = (K_{\text{SM}} - 1) + C_{\text{vacuum}} = 60 + 42 = 102$ at the SM interface. The 42 appearing in the numerator is C_{vacuum}, the vacuum-residual per-slot admissibility count. The 102 in the denominator is the total per-slot admissibility under self-exclusion. Their ratio is therefore the fraction of per-slot admissibility allocated to the vacuum-residual sector; both numerator and denominator derive from the same bank theorem.

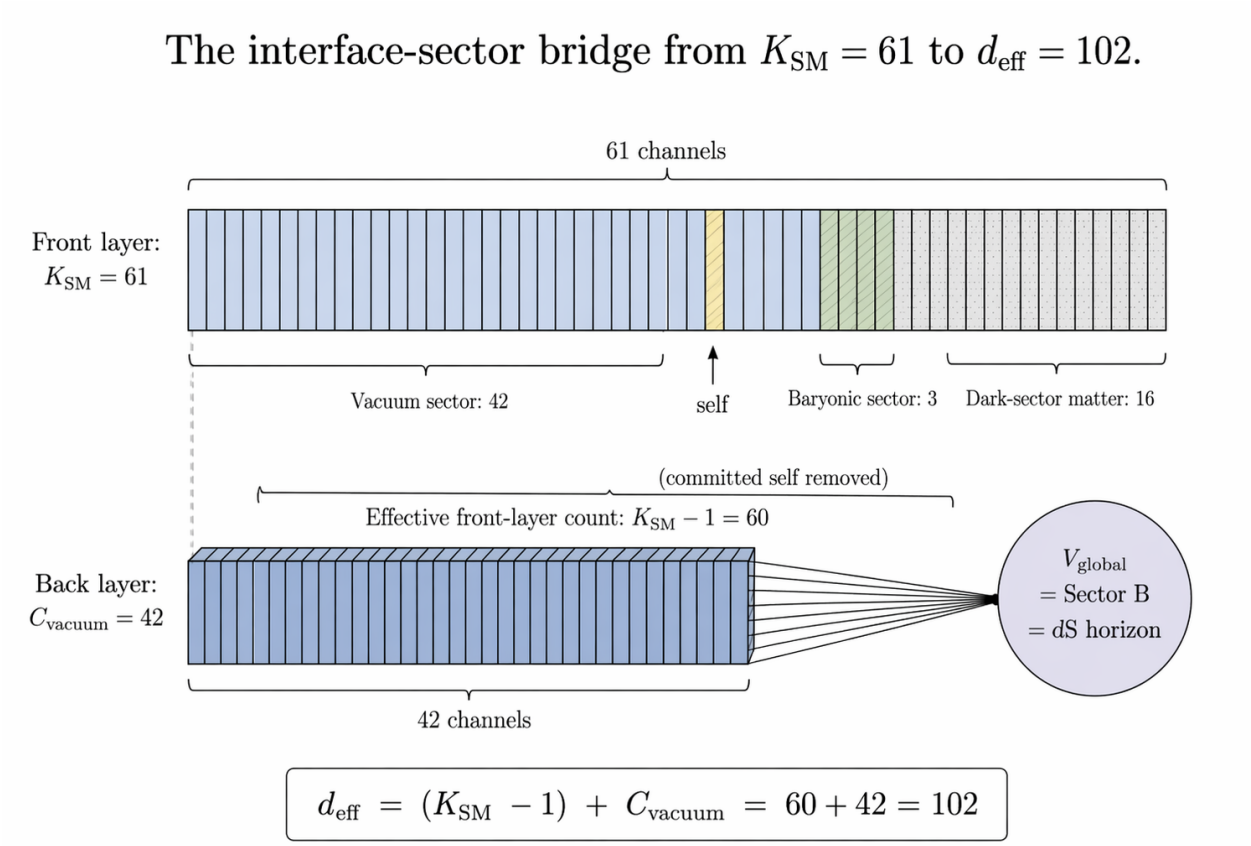

Figure 3: The interface-sector bridge from $K_{\mathrm{SM}} = 61$ to $d_{\mathrm{eff}} = 102$. The front layer (top bar) is the 61-channel SM budget of Fig. 1, with one arbitrary channel committed as "self" (yellow) and therefore removed from its own count; the effective front-layer population is $K_{\mathrm{SM}} - 1 = 60$ (Sector A of $T_{\mathrm{horizon_reciprocity}}$). The back layer (lower bar) is the 42-dimensional vacuum-residual stratum V_{global} — the same 42 vacuum channels shown in blue on the front layer, now read as Sector B and identified with the de Sitter horizon absorber via $L_{\mathrm{global_interface_is_horizon}}$ (*Paper 6* ()). The per-slot effective admissibility decomposes as $d_{\mathrm{eff}} = (K_{\mathrm{SM}} - 1) + C_{\mathrm{vacuum}} = 60 + 42 = 102$. The 42/102 coefficient in the dark-energy absolute prediction is the ratio of the back-layer count to the total.

A numerical coefficient scan over APF-native arithmetic ratios (ratios of bank-forced integers with no mathematical-constant prefactors) within 0.01 decades of the observed $\rho_\Lambda/M_{\mathrm{Pl}}^4$ finds additional candidates at comparable numerical precision, including 19/45, 42/45, and 41/45. These are arithmetic expressions of bank integers, but they do not correspond to derivation chains in the bank terminating at a vacuum-fraction identification: in 19/45 the numerator $C_{\mathrm{local}} = 19$ is the finite-interface stratum dimension, not the vacuum-residual count; in 42/45 and 41/45 the denominators are the total fermion slot count K_{fermions}, not the per-slot admissibility. Of the APF-native candidates at observational precision, $42/102 = C_{\mathrm{vacuum}}/d_{\mathrm{eff}}^{\mathrm{SM}}$ is the ratio whose derivation chain terminates at $L_{\mathrm{self_exclusion}}$'s vacuum-fraction structure, and the 42 in the prediction is specifically the C_{vacuum} that V_Λ's dimension equals at the SM interface.

Code reference: `check_T_Lambda_coefficient_degeneracy_audit` in `apf/lambda_absolute.py`.

8.3 Absolute predictions for the cosmological energy budget

Applying the formula to each cosmological density component:

$$\begin{aligned}
\rho_{\rm crit}/M_{\rm Pl}^4 &= 61/102^{62} = 10^{-122.748}, \\
\rho_b/M_{\rm Pl}^4 &= 3/102^{62} = 10^{-124.056}, \\
\rho_c/M_{\rm Pl}^4 &= 16/102^{62} = 10^{-123.329}, \\
\rho_\Lambda/M_{\rm Pl}^4 &= 42/102^{62} = 10^{-122.910}.
\end{aligned}$$

The dark-energy prediction deviates from the Planck-2018-inferred value of $\rho_\Lambda/M_{\rm Pl}^4 = 10^{-122.944}$ [1] by approximately 0.034 decades, corresponding to an 8% ratio. This deviation is of the same order as the current Hubble tension: the observational value of $\rho_\Lambda/M_{\rm Pl}^4$ depends on which choice of H_0 is used, and the Planck-2018 and SH0ES-2022 values bracket a range spanning comparable magnitude.

Code reference: `check_T_Lambda_absolute_extended_formula` in `apf/lambda_absolute.py`.

8.4 H_0 from the algebraic inversion

The dark-energy absolute prediction, combined with the bank-forced $\Omega_\Lambda = 42/61$ and the standard GR critical-density formula $\rho_{\rm crit} = 3H_0^2 M_{\rm Pl}^2/(8\pi)$, determines H_0 algebraically. Solving:

$$\rho_\Lambda = \Omega_\Lambda \cdot \rho_{\rm crit} \implies \frac{\rho_\Lambda}{M_{\rm Pl}^4} = \Omega_\Lambda \cdot \frac{3}{8\pi}\left(\frac{H_0}{M_{\rm Pl}}\right)^2,$$

$$\left(\frac{H_0}{M_{\rm Pl}}\right)^2 = \frac{8\pi}{3} \cdot \frac{\rho_\Lambda/M_{\rm Pl}^4}{\Omega_\Lambda} = \frac{8\pi}{3} \cdot \frac{42/102^{62}}{42/61} = \frac{8\pi \cdot 61}{3 \cdot 102^{62}}.$$

APF prediction for the Hubble constant

$$\frac{H_0}{M_{\rm Pl}} = \frac{\sqrt{8\pi K_{\rm SM}/3}}{102^{(K_{\rm SM}+1)/2}} = \frac{\sqrt{8\pi \cdot 61/3}}{102^{31}} = 1.224 \times 10^{-61}.$$

In SI units, $H_0 \approx 70.03\,{\rm km\,s^{-1}\,Mpc^{-1}}$.

Code reference: `check_T_Lambda_to_H0_inversion` in `apf/lambda_absolute.py`.

The SI-form Hubble constant prediction lies between the Planck-2018 value $H_0^{\rm Planck} = 67.36\,{\rm km\,s^{-1}\,Mpc^{-1}}$ [1] and the SH0ES-2022 value $H_0^{\rm SH0ES} = 73.04\,{\rm km\,s^{-1}\,Mpc^{-1}}$ [2], at 0.17 km/s/Mpc below the present tension midpoint. The 8% deviation of the Paper-8 ρ_Λ prediction from Planck-2018's inferred value corresponds to the 4% deviation of the Paper-8 H_0 prediction from the Planck-2018 H_0, and the same paired deviations with opposite signs against SH0ES 2022. The ρ_Λ prediction and the H_0 prediction are algebraically the same prediction read on the two sides of the standard critical-density relation.

8.5 The CMB temperature, independent observable

The cosmic microwave background temperature is an observable that does not factor through the matter-sector prediction. Its APF prediction follows a species-dependent analog of the matter-sector formula, with exponent $k = K_{\mathrm{SM}} + 3 = 64$ rather than $K_{\mathrm{SM}} + 1 = 62$:

CMB thermal-bath formula

$$\left(\frac{T_{\mathrm{CMB}}}{M_{\mathrm{Pl}}}\right)^4 = \frac{C_T}{d_{\mathrm{eff}}^{K_{\mathrm{SM}}+3}} = \frac{48}{102^{64}},$$

with the coefficient $C_T = 48$ equal to $C_c \times C_b = K_{\mathrm{gauge}} \times K_{\mathrm{Higgs}}$ (both readings bank-forced and numerically equal) and the exponent shift $\Delta k = +2$ matching the two photon polarization states. In SI units, the predicted temperature is $T_{\mathrm{CMB}}^{\mathrm{APF}} = 2.7166\,\mathrm{K}$; the FIRAS-measured value is $T_{\mathrm{CMB}}^{\mathrm{FIRAS}} = 2.7255 \pm 0.0006\,\mathrm{K}$ [3], a deviation of 0.33%.

Code reference: `check_T_T_CMB_absolute_formula` in `apf/thermal_absolute.py`.

The matter-sector exponent $K_{\mathrm{SM}} + 1 = 62$ and the photon-sector exponent $K_{\mathrm{SM}} + 3 = 64$ differ by $\Delta k = 2$, matching the two photon polarisation states. This motivates a species-dependent hypothesis $k_X = K_{\mathrm{SM}} + 1 + N_{\mathrm{pol,X}}$ — registered [C, open] pending additional thermal-bath data points. Full construction, remaining assumptions, and what would close the status tag: Supplement §J.

8.6 Bulletproofing and scope

The six cosmological predictions (four matter-sector densities plus H_0 plus T_{CMB}) are internally consistent and individually verifiable by code check. Their composed bulletproof theorem is:

Theorem T_Lambda_absolute_bulletproof

The matter-sector formula $\rho_X/M_{\mathrm{Pl}}^4 = C_X/102^{62}$ gives four cosmological-density predictions $(\rho_{\mathrm{crit}}, \rho_b, \rho_c, \rho_\Lambda)$ at a common $\approx 8\%$ deviation from Planck-2018 cosmology [1]; the algebraically-forced $H_0 = 70.03\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ falls between Planck 2018 [1] and SH0ES 2022 [2], near the present tension midpoint; the CMB thermal-bath formula $(T_{\mathrm{CMB}}/M_{\mathrm{Pl}})^4 = 48/102^{64}$ with species-dependent exponent $k = K_{\mathrm{SM}} + 3$ matches FIRAS [3] at 0.33%. All six predictions use only bank-forced structural constants (K_{SM}, $d_{\mathrm{eff}}^{\mathrm{SM}}$, C_b, C_c, C_{vacuum}, C_T) and the Planck-unit convention of §1.

Code reference: `check_T_Lambda_absolute_bulletproof` in `apf/lambda_absolute.py`.

§9 shows that this derivation chain admits an operator-level realization on an explicit finite-dimensional microstate Hilbert space. §10 delineates what the prediction formula covers and what it does not.

9. Operator-Level Realization (Finite-Dimensional)

Scope and non-claims. The operator-level construction below is *finite-dimensional by design* and is used exclusively in the $\beta \to 0$ high-temperature limit. No claims are made about thermodynamic-limit existence ($N \to \infty$, $K \to \infty$), about C^*- or von Neumann algebraic structure beyond the finite $M_N(\mathbb{C})$ realisation, about the KMS condition at non-zero β, or about von Neumann factor types in the infinite-volume limit. The construction's role in Paper 8 is to supply an operator-level witness for the I_4 self-identification and the matter-sector cosmological formula $\rho_X/M_{\mathrm{Pl}}^4 = C_X/d_{\mathrm{eff}}^{K+1}$, not to engage with algebraic quantum statistical mechanics. Infinite-volume extension would require importing the standard apparatus (quasi-local C^*-algebras, KMS states, modular theory) and is deferred; see Supplement §H.1 for what this extension would require, and Supplement §N.3 for the positioning relative to algebraic QSM.

This section constructs an operator-level realization of the ledger identities proved in §2–§6; it is not a second derivation from independent premises, but the concrete operator-algebra form of the same bookkeeping. The construction gives an explicit finite-dimensional microstate Hilbert space, defines a thermal density matrix on it, and recovers the cosmological predictions of §8 as $\beta \to 0$ limits of operator-algebra identities.

9.1 The thermal construction

Fix an ACC interface record (K, d_{eff}) with C_{vacuum} per-slot vacuum-residual states and $(d_{\mathrm{eff}} - C_{\mathrm{vacuum}})$ per-slot excited states. The microstate Hilbert space is $H_{\mathrm{micro}} = (\mathbb{C}^{d_{\mathrm{eff}}})^{\otimes K}$ of dimension $N = d_{\mathrm{eff}}^K$; the total Hamiltonian is the sum of local vacuum-versus-excited Hamiltonians acting factor-wise; and the thermal density matrix is $\rho_\beta = Z(\beta)^{-1} e^{-\beta H_{\mathrm{total}}}$. At the $\beta \to 0$ limit, $\rho_{\beta\to 0} = N^{-1}\mathbf{1}$ is the maximally mixed density matrix on H_{micro}. Full construction details (local Hamiltonian eigenstructure, vacuum projector $P_{\mathrm{vac},i}$, trace normalisations) are in Supplement §H.

Code reference: `_build_model` in `apf/lambda_operator_derivation.py`.

9.2 Three operator-algebra identities at $\beta \to 0$

At the $\beta \to 0$ limit of the thermal construction, three structural identities hold to machine precision at every test interface. Each is a rigorous operator-algebra statement, not an asymptotic approximation; each is I_4's scalar witness (A), the per-slot vacuum fraction (B), and the admissibility-suppression factor (C) of §8 respectively.

Identity (A) — Partition-function limit

$$\ln Z(\beta \to 0) \;=\; \ln N \;=\; K \ln d_{\mathrm{eff}} \;=\; \text{ACC-scalar}.$$

Identity (B) — Vacuum-projector expectation

$$\langle P_{\mathrm{vac},i}\rangle_{\rho_{\beta\to 0}} = \frac{\mathrm{Tr}(P_{\mathrm{vac},i})}{N} = \frac{C_{\mathrm{vacuum}}\cdot d_{\mathrm{eff}}^{K-1}}{d_{\mathrm{eff}}^{K}} = \frac{C_{\mathrm{vacuum}}}{d_{\mathrm{eff}}}.$$

Identity (C) — Per-microstate probability

$$p_{\mathrm{microstate}}(\rho_{\beta\to 0}) = \frac{1}{N} = \frac{1}{d_{\mathrm{eff}}^{K}}.$$

At the SM interface, (B) evaluates to 42/102 and (C) to 1/102[61], the two coefficients of the matter-sector formula. Supplement §H gives the derivation of each identity, the operational content of each on the thermal state, and the step-by-step composition showing how (A), (B), (C) together yield $\rho_\Lambda/M_{\mathrm{Pl}}^4 = 42/102$[62].

Code reference: `check_T_Lambda_partition_function_at_beta_zero` in `apf/lambda_operator_derivation.py`.

9.3 The composed operator-level theorem

Theorem T_Lambda_d2_operator_derivation

The matter-sector cosmological absolute formula $\rho_X/M_{\mathrm{Pl}}^4 = C_X/d_{\mathrm{eff}}^{K+1}$ of §8 admits an operator-level realisation on $H_{\mathrm{micro}} = (\mathbb{C}^{d_{\mathrm{eff}}})^{\otimes K}$. The three $\beta \to 0$ identities (A), (B), (C) are rigorous operator-algebra statements verified to machine precision at a representative family of test interfaces; their composition, with the Planck-units convention of §1, reproduces the matter-sector formula. Full proof and operational verification: Supplement §H.

Code reference: `check_T_Lambda_d2_operator_derivation` in `apf/lambda_operator_derivation.py`.

9.4 Operator-level composed self-identifications for I_3 and I_4 (Phase 14f.1 + 14f.3)

The three operator-algebra identities (A), (B), (C) of §9.2 deliver more than the matter-sector composition of the composed theorem above — they also supply the structurally distinct readings that close I_3 and I_4 at the subspace level as *composed self-identifications*, paralleling the role $T_{\mathrm{interface_sector_bridge}}$ plays for I_2.

Theorem T_I3_operator_self_identification

The I_3 (thermo-quantum) identity is closed by a three-construction composed self-identification on the quantum-interface carrier $\mathbb{C}^d$: direct numpy $\rho_{\max} = (1/d)I_d$ with $S_{\mathrm{vN}} = \ln d$; the Schur-lemma characterisation of $\rho_{\max}$ as the unique $U(d)$-invariant normalised state; and the $K = 1$ specialisation of the thermal machinery above, in which $\rho_\beta \to \rho_{\max}$ as $\beta \to 0$ for any local Hamiltonian. All three deliver the same density matrix.

Status [$\mathrm{P_{comp}}$].

Code reference: `check_T_I3_operator_self_identification` in `apf/quantum_operator_derivation.py`.

Theorem T_I4_operator_self_identification

The I_4 (action-thermo) identity is closed by a four-reading composed self-identification on H_{micro}: identities (A), (B), (C) above together with (D) Paper 7's spectral-action reading $S_{\mathrm{spec}} = \mathrm{Tr}[\exp(-\beta D^2)]$, which evaluates to the same $\ln Z = \mathrm{ACC}_{\mathrm{scalar}}$ at $\beta \to 0$ via the heat-kernel expansion's leading coefficient $a_0 = \dim H_F$. All four readings recover the same $\mathrm{ACC}_{\mathrm{scalar}}$. Status [$\mathrm{P_{comp}}$].

Code reference: `check_T_I4_operator_self_identification` in `apf/i4_composition.py`.

The closure species at I_3 and I_4 is single-interface composed self-identification (not bridge), because at `acc_quantum`(d) the slot scale $K = 1$ collapses and there is no companion interface to bridge to. This is structurally parallel to how I_1's bridge theorem is available only at the $K = 42$ joint point (where both interfaces carry non-trivial slot content), while generic horizons close regime-locally. The four-way stratification — two bridges (I_2 always, I_1 at $K = 42$), two composed self-identifications (I_3, I_4) — is the Phase 14f-final form of the T_{ACC} subspace closure. Technical Supplement §E.6 is the catalogue.

The operator-level realisation is the concrete operator-algebra form of the bookkeeping that §§2–6 assemble. §10 now delineates the scope over which the formula applies.

10. Robustness and Scope

The cosmological formula of §8 has a specific domain of applicability. This section delineates what the formula covers, what it does not, and what open structural questions remain about the extensions.

10.1 In-scope observables

The matter-sector formula $\rho_X/M_{\mathrm{Pl}}^4 = C_X/d_{\mathrm{eff}}^{K_{\mathrm{SM}}+1}$ covers the four cosmological-density components of a flat universe at the present epoch ($\rho_{\mathrm{crit}}, \rho_b, \rho_c, \rho_\Lambda$), with slot-count inputs $C_X \in \{K_{\mathrm{SM}}, C_b, C_c, C_{\mathrm{vacuum}}\} = \{61, 3, 16, 42\}$ all bank-forced. The thermal-bath formula $(T_X/M_{\mathrm{Pl}})^4 = C_T/d_{\mathrm{eff}}^{K_{\mathrm{SM}}+3}$ covers photon thermal backgrounds at the present epoch, with coefficient $C_T = 48$ bank-forced. The species-dependent exponent hypothesis $k_X = K_{\mathrm{SM}} + 1 + N_{\mathrm{pol,X}}$ extends in principle to additional thermal-bath observables at predicted exponents pending data-point confirmation.

10.2 Open extensions: neutrino mass sum and the cosmic neutrino background

Two cosmological observables lie on the edge of the formula's current reach but are not yet closed: the *neutrino mass sum* Σm_ν, which admits a suggestive numerical fit $\sim 11/102^{15}$ within the observational window, and the *cosmic neutrino background*, whose Weyl-neutrino polarisation count

$N_{\text{pol}} = 1$ would test the species-dependent exponent hypothesis at $k = K_{\text{SM}} + 2 = 63$. Both are registered [C, open] pending improved measurement precision. Full discussion of the Σm_ν fit's bank-native readings and what would close its status, together with the CνB prediction's assumptions and accessibility, is in Supplement §J.

Code reference: `check_L_Sigma_m_nu_suggestive` in `apf/thermal_absolute.py`.

10.3 Out of scope: the baryon-to-photon ratio

The baryon-to-photon ratio $\eta \approx 6.1 \times 10^{-10}$ does not admit a clean APF-native C/d_{eff}^k fit at any small integer exponent. Numerical scans return arithmetic candidates without derivation chains terminating in bank structure. The observed value reflects baryogenesis physics and the specific CP-violation content of the Standard Model, neither of which is captured by the two-tier admissibility-capacity formula of §8. This does not contradict APF; it marks the boundary where the cosmological-sector prediction formula ceases to apply.

Code reference: `check_L_eta_does_not_fit_cleanly` in `apf/thermal_absolute.py`.

10.4 Scope summary

Table 9 summarizes the current scope of the prediction formula across observable types.

Observable	**Status**	**Note**
$\rho_\Lambda, \rho_b, \rho_c, \rho_{\text{crit}}$	Covered	Matter-sector formula, $k = K_{\text{SM}} + 1 = 62$
T_{CMB}, ρ_γ	Covered	Thermal-bath formula, $k = K_{\text{SM}} + 3 = 64$
H_0	Covered	Algebraic inversion from $\rho_\Lambda + \Omega_\Lambda$ + GR critical density
Σm_ν	[C, open]	Suggestive fit consistent with observational window; coefficient not structurally pinned
Cosmic ν-background	Predicted	Species-dependent exponent hypothesis at $N_{\text{pol}} = 1$: $k = 63$
η (baryon-to-photon ratio)	Out-of-scope	Baryogenesis physics; no bank derivation chain terminates at a C/d_{eff}^k form

Table 9: Scope of the Paper-8 prediction formula across standard cosmological observables.

11. Falsifiers

The paper's quantitative claims rest on bank-forced structural constants and one formula with a species-dependent exponent. Each component is falsifiable by explicit target.

(F1) Hubble-constant convergence

If future Hubble-constant measurements converge to either the Planck-2018 value ($67.36\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$ [1]) or the SH0ES-2022 value ($73.04\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$ [2]) to within their current uncertainties, with the other value excluded, the Paper-8 prediction $H_0 = 70.03\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$ is falsified at approximately 4%. The prediction commits APF to the present tension midpoint; resolution in favor of either extreme falsifies.

(F2) C_{vacuum} rederivation

If independent rederivation of Paper 6's T12 partition gives $C_{\text{vacuum}} \neq 42$, the dark-energy numerator shifts and the $\rho_\Lambda/M_{\text{Pl}}^4 = 42/102^{62}$ prediction is falsified. Independent rederivation from distinct structural machinery — spectral-action, NCG, or observer-dependent reformulation — is a target.

(F3) K_{SM} rederivation

If independent rederivation of `L_count` (Paper 2) gives $K_{\text{SM}} \neq 61$, every paper-level formula shifts simultaneously: the exponent changes, the Ω_X denominators change, the H_0 prediction changes. $K_{\text{SM}} = 61$ is the single most load-bearing integer in Paper 8.

(F4) d_{eff} rederivation

If independent rederivation of `L_self_exclusion` (Paper 6) gives $d_{\text{eff}}^{\text{SM}} \neq 102$, the per-slot admissibility base shifts and every formula in this paper moves. $d_{\text{eff}} = 102 = 60 + 42$ is a structural sum of two numerically independent quantities ($K_{\text{SM}} - 1$ and C_{vacuum}); an independent rederivation of either factor is a target.

(F5) CMB temperature and species-dependent exponent

If the species-dependent exponent hypothesis $k_X = K_{\text{SM}} + 1 + N_{\text{pol,X}}$ fails at a cleanly-measurable species beyond the matter ($k = 62$) and photon ($k = 64$) confirmations, the hypothesis is falsified as a structural rule. A candidate independent test is the thermal-photon prediction's continued match to FIRAS [3] at the 0.33% level; larger discrepancies under improved measurement or reanalysis would downgrade the thermal formula.

(F6) 42/102 uniqueness

If the coefficient-uniqueness argument of §8.2 fails under reanalysis — specifically, if an alternative APF-native ratio is shown to have a derivation chain terminating at a vacuum-fraction reading comparable to $C_{\text{vacuum}}/d_{\text{eff}}^{\text{SM}}$ — the 42/102 coefficient's structural privilege is undermined. The claim rests on the bank state as of this paper; a rederivation that legitimizes one of the arithmetic lookalikes would shift the coefficient.

12. Discussion

The integrated non-trivial content lives in the centerpiece Supplement §O. The structural mechanism of Paper 8 — the I_2 bridge, the V_Λ gauge-uniqueness (Supplement Theorem 5.7, v2.3), the framework-internal forcing chain yielding $(K_{\text{SM}}, d_{\text{eff}}^{\text{SM}}) = (61, 102)$ (Supplement Appendix M), the two-tier single-scale impossibility theorem (Supplement Theorem C.5, v2.3), the brittle-

consequences sensitivity calculation, and the multivariate Planck 2018 Bayes-factor confrontation with $\chi^2 = 1.18$ over three observables (Supplement §I′.4) — is collected in one focal supplement section: §O, "The Gauge-Cosmological Bridge: Integrated Nontrivial Content." A reader wanting the headline structural content of the paper should read that section first; the detailed technical material lives in the surrounding supplement sections. Readers wanting the algebraic-QSM content should note that v2.3 also closes the infinite-volume factor-type question (Supplement Theorem 7.14′: the thermodynamic-limit von Neumann algebra is Type III_1 at every finite β, via Araki–Connes classification).

Paper 8's quantitative results depend on the SM-interface slot-subspace V_Λ being structurally well-defined. The bridge theorem of *Paper 6* () and the matter-sector formula of §8 collectively give V_Λ three identifications:

1. V_Λ as the global-interface stratum of the T12 partition of V_{61} (observer-independent in APF; algebraic).
2. V_Λ as Sector B of the second-ε decomposition of $T_{\text{horizon-reciprocity}}$ (observer-independent in APF; algebraic).
3. V_Λ as the de Sitter horizon absorber subspace at the $K = 42$ joint SM–dS point (observer-dependent in general relativity; cosmological horizons depend on observer worldlines).

The first two are observer-independent; the third is observer-dependent as it is normally formulated in GR. Three logically possible resolutions of the mismatch: (a) the observer-dependence is trivial at the $K = 42$ joint point and the three identifications coincide by structural accident there; (b) the bridge theorem implicitly fixes a comoving de Sitter observer, and the identification holds only in that frame; (c) V_Λ is genuinely observer-invariant as an ACC-interface structural object, and the apparent observer-dependence in GR is an artifact of metric-framing that dissolves under a categorical reformulation of the bridge theorem. Resolution (c) would make APF's cosmological-sector content a structural prediction beyond ΛCDM: different observers disagree on local horizon metrics but agree on the 42-dimensional V_Λ content of any cosmological-scale horizon they construct.

The quantitative claims of this paper do not depend on which resolution is correct. $\Omega_\Lambda = 42/61$, $\rho_\Lambda/M_{\mathrm{Pl}}^4 = 42/102^{62}$, H_0 from the algebraic inversion, and the CMB formula all follow from the SM-interface bank machinery before the observer-dependence question is raised. The question is registered as the paper's one substantive open structural question; categorical-reformulation work is a natural next direction in the series.

Code reference: `check_T_bridge_observer_independence_open` in `apf/phase_14d3_completions.py`.

Distinct but orthogonal question: horizon parametrisation. A second ambiguity, unrelated to the observer-dependence question above, concerned the *parametrisation* of the horizon interface: APF's default Convention A takes $d_{\text{eff}} = e$ at horizons (so $\mathrm{ACC}_{\text{horizon}} = K$ matches S_{BH} in nats directly), but this choice makes the horizon Hilbert space of dimension e^K schematic rather than literal — no integer-dimensional Hilbert space has dimension e. Phase 14f.2 (lemma $L_{\text{horizon_convention_equivalence}}$, `unification.py`) formalises the alternative Convention B ($d_{\text{eff}} = 2$, cell size $4 \ln 2\, \ell_P^2$, $K = K_{\text{bit}}$ integer bit-cells, $\dim H_{\text{micro}} = 2^{K_{\text{bit}}}$ a literal qubit tensor product) and proves the two parametrisations deliver the same $\mathrm{ACC}_{\text{scalar}}$ and the same ambient Hilbert dimension at every horizon. Slot-scale

identifications (absorber subspaces, V_{global}, the $K = 42$ SM-dS joint point) are convention-invariant; the dual-convention formulation resolves the parametrisation ambiguity by accepting both readings and using whichever is natural per use case. The I_1 joint-point bridge theorem's Construction 3 (§5) uses Convention B to deliver the literal $(\mathbb{C}^2)^{\otimes 42}$ qubit realisation that Convention A cannot. Technical Supplement §K.4 records the resolution.

Code reference: `check_L_horizon_convention_equivalence` in `apf/unification.py`.

13. Conclusion

The Admissibility-Capacity Ledger, assembled in Papers 1–7 and summarized in *Paper 13* (), closes at the cosmological sector. The three-level identity stack of §4 has a subspace-level witness [P] at every T_ACC consistency identity, supplied by the four subspace functors of §5; the Fractional Reading Equivalence of §6 identifies the Standard-Model cosmological residual fractions with information-entropy allocations on the V_{61} slot-space; the matter-sector absolute formula of §8 gives dimensionless predictions in Planck units for ρ_Λ, ρ_{crit}, ρ_b, ρ_c, the algebraically-linked H_0, and (via the species-dependent exponent hypothesis) the CMB thermal-bath temperature; the operator-level realization of §9 supplies the same derivations on an explicit finite-dimensional microstate Hilbert space. The paper's cosmological content is therefore the direct application of the already-established ledger machinery to the cosmological sector.

One axiom (**A1**, Finite Enforcement Capacity). No imported physics structure: the Standard-Model gauge group is derived in *Paper 4* (), not assumed; the generation count is derived in *Paper 2* (), not assumed; the weak mixing angle $\sin^2\theta_W = 3/13$ is derived in *Paper 4* (), not fitted; the cosmological residual fractions $\{3/61, 16/61, 42/61\}$ are derived here, not chosen. Mathematical theorems used in closure and representation arguments (Gleason, Stinespring, Lieb–Ruskai, Tarjan 1972, and others) are cited at point of use as explicit formal mathematical imports, not hidden physics assumptions. Zero free parameters.

Planck units are APF's native unit system: every derivation produces a dimensionless number or a dimensionless ratio times M_{Pl}^n for integer n, with $M_{\text{Pl}} = 1$ by convention. Laboratory comparison with SI-unit experiments uses one external input, Newton's gravitational constant $G = M_{\text{Pl}}^{-2}$, at the comparison stage only. This is the same unit-interface practice used by quantum field theory, lattice QCD, and every other natural-units framework.

The quantitative scoreboard: $\Omega_b, \Omega_c, \Omega_\Lambda$ predicted as entropy fractions matching Planck-2018 [1] to fractions of a percent; $\rho_\Lambda/M_{\text{Pl}}^4 = 42/102^{62}$ deviating from the Planck-2018-inferred value by an 8% margin of the same order as the current Hubble tension; the algebraically-linked $H_0 = 70.03\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$ landing 0.17 km/s/Mpc from the present tension midpoint between Planck 2018 and SH0ES 2022 [2]; T_{CMB} predicted by the independent thermal-bath formula matching FIRAS [3] to 0.33%.

The substantive open structural question raised by the paper's content is the observer-dependence of the bridge theorem as applied to V_Λ. The quantitative cosmological predictions do not depend on its resolution; a categorical reformulation of the bridge that resolves it is the natural next direction in the series.

References

[1] Planck Collaboration (N. Aghanim *et al.*), "Planck 2018 results. VI. Cosmological parameters," *Astronomy & Astrophysics* **641**, A6 (2020). arXiv:1807.06209.

[2] A. G. Riess *et al.* (SH0ES team), "A Comprehensive Measurement of the Local Value of the Hubble Constant with $1\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ Uncertainty from the Hubble Space Telescope and the SH0ES Team," *Astrophysical Journal Letters* **934**, L7 (2022). arXiv:2112.04510.

[3] D. J. Fixsen, "The Temperature of the Cosmic Microwave Background," *Astrophysical Journal* **707**, 916 (2009). arXiv:0911.1955.

[4] R. L. Workman *et al.* (Particle Data Group), "Review of Particle Physics," *Progress of Theoretical and Experimental Physics* **2022**, 083C01 (2022) and 2024 update. pdg.lbl.gov.

[5] E. S. Brooke, "The Weak Mixing Angle as a Capacity Equilibrium," Zenodo (2026). doi:10.5281/zenodo.18603209.